

Calculus on Manifolds

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(Based on the lecture notes of Professor Florian Naef)

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"We must... maintain that mathematical geometry is not a science of space insofar as we understand by space a visual structure that can be filled with objects - it is a pure theory of manifolds."

-Hans Reichenbach

The following is based on the course "Calculus on Manifolds" as it was taught by Professor Florian Naef. Although the notes contain everything provided in the lecture notes, I would say that these should not serve as a substitute for attending lectures or learning from a proper textbook ("Analysis On Manifolds" By Munkres and "Introduction to Smooth Manifolds" by Lee are the books that most align with the content of the course.) Manifolds are fundamentally geometric objects, pictures are hard to do in LaTeX but it's a good learning exercise to try and draw diagrams of what's going on for yourself. Both books named above have an abundance of diagrams.

A lot of results are left without proof, unless it's explicitly stated that the proof is omitted, the proof should be taken as an exercise (of wildly varying difficulty) for the reader.

Although I've gone through several drafts, mistakes are bound to have slipped through so if anything seems illogical, it's (probably) a mistake on my part so you should cross check with other sources. Notation in this subject can be cumbersome so it might be good to make a glossary of functions, definitions, symbols etc.

I would greatly appreciate any feedback/corrections. You can email me at fleminev@tcd.ie

Happy Manifolding!

Chapter 1

Review of Topology and Real Analysis

1.1 Metric Spaces

Definition 1.1.1 (Metrics and Metric Spaces). A metric on a set X is a function $d : X \times X \rightarrow \mathbb{R}$ such that the following conditions hold $\forall x, y, z \in X$:

1. $d(x, y) = d(y, x)$
2. $d(x, y) = 0 \iff x = y$
3. $d(x, y) \leq d(x, z) + d(z, y)$

The pair (X, d) is a metric space. For example, $(\mathbb{R}^n, d_2)$ where $d_2(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$ is a metric space.

Definition 1.1.2 (Open and Closed sets). A set $U \subset X$ is open if and only if $\forall x \in U, \exists \varepsilon > 0$ such that $B(x, \varepsilon) \subset U$ where $B(x, \varepsilon) = \{p \in X \mid d(x, p) < \varepsilon\}$. A set is closed if its complement is open. We shall denote the set of open sets in any space (X, d) as $\mathcal{T}_X$.

Proposition 1.1.3. Given $U \subset Y \subset X$ where (X, d) is a metric space, U being open/closed in $Y \iff \exists V \subset X$ (where V is open/closed in X) such that $U = V \cap Y$

Definition 1.1.4 (Continuity). A function $f : X \rightarrow Y$ is continuous if and only if the preimage of every open set in Y is open in X .

Definition 1.1.5 (Homeomorphism). if $f : X \rightarrow Y$ is a continuous bijection and f^{-1} is continuous as well, then f is a homeomorphism. If there exists a homeomorphism between sets, they're homeomorphic.

Definition 1.1.6 (Compactness). X is compact if and only if every open cover of X has a finite sub-cover.

Theorem 1.1.7 (Heine-Borel). Compact in $\mathbb{R}^n \iff$ closed and bounded.

1.2 Differentiation

Definition 1.2.1 (Differentiability). If $\mathcal{U} \subset \mathbb{R}^n$, $\mathcal{V} \subset \mathbb{R}^m$, with $f : \mathcal{U} \rightarrow \mathcal{V}$, f is differentiable at $\mathbf{a} \in \mathcal{U}$ with derivative $Df(\mathbf{a})$ (as shown below) $\in \mathcal{L}(\mathcal{U}, \mathcal{V})$ if and only if the following limit holds

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} \frac{|f(\mathbf{x}) - f(\mathbf{a}) - Df(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a})|}{|\mathbf{x} - \mathbf{a}|} = 0$$

Definition 1.2.2 (Functions of class C^r and smooth functions). f is of class C^1 if it is differentiable at all points in U and $Df : \mathcal{U} \rightarrow \text{Mat}_{m \times n}$ is continuous at all points. f is of class C^r if Df is of class C^{r-1} . f is smooth or of class C^∞ if it is C^r for all positive integers r .

Proposition 1.2.3. $f : \mathcal{U} \rightarrow \mathbb{R}^m$ is C^1 if and only if $\frac{\partial}{\partial x_j}(f_i) : \mathcal{U} \rightarrow \mathbb{R}$ exists and is continuous for every i, j . f is C^r if $\frac{\partial^r}{\partial x_i^r}(f_i) : \mathcal{U} \rightarrow \mathbb{R}$ exist and are continuous for every i, j .

$$Df(\mathbf{a}) = \begin{bmatrix} \frac{\partial f_1(\mathbf{a})}{\partial x_1} & \dots & \frac{\partial f_1(\mathbf{a})}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(\mathbf{a})}{\partial x_1} & \dots & \frac{\partial f_m(\mathbf{a})}{\partial x_n} \end{bmatrix}$$

Theorem 1.2.4 (Multivariable Chain Rule). if $f : \mathcal{U} \rightarrow \mathcal{V}$ and $g : \mathcal{V} \rightarrow \mathcal{W}$ are differentiable/ C^r , then their composition is differentiable/ C^r and it's derivative is given by the formula:

$$D(g \circ f)(\mathbf{a}) = Dg(f(\mathbf{a})) \circ Df(\mathbf{a})$$

Definition 1.2.5 (C^r Diffeomorphism). If $f : \mathcal{U} \rightarrow \mathcal{V}$ with U, V open, is of class C^r and its inverse exists and is also of class C^r , f is a C^r diffeomorphism.

Theorem 1.2.6 (Inverse Function Theorem). If $\mathcal{U} \subset \mathbb{R}^n$ and $f : \mathcal{U} \rightarrow \mathbb{R}^n$ is of class C^r and $Df(\mathbf{a})$ is invertible for some $\mathbf{a} \in \mathcal{U}$, then there exists some open subset V of U , such that if f is restricted to V , f is a diffeomorphism i.e $f|_V : V \rightarrow f(V)$ is a C^r diffeomorphism.

Chapter 2

Manifolds in $\mathbb{R}^n$

2.1 Definition of a Manifold in $\mathbb{R}^n$

Definition 2.1.1 (Smooth/ C^r Manifold). Let $d > 0$. $M \subset \mathbb{R}^n$ is a smooth/ C^r manifold of dimension d if $\forall p \in M \exists V \in \mathcal{T}_{\mathbb{R}^n}, V \subset M, U \in \mathcal{T}_{\mathbb{R}^d}$ and a function $\alpha : U \rightarrow V$ where α is:

1. Smooth/ C^r
2. Is a bijection with a continuous inverse (a homeomorphism)
3. $D\alpha(x)$ has rank $d \forall x \in U$. This means all d of its columns are linearly independent.

We call α a coordinate patch around p . We often refer to the pair (α, U) as a chart. Essentially what all this is saying is that M is a manifold of dimension d if each point has a neighbourhood that is homeomorphic to an open set in $\mathbb{R}^d$. In $\mathbb{R}^n$ M is a manifold, roughly speaking, if it has a smooth shape without self-intersection, corners or creases.

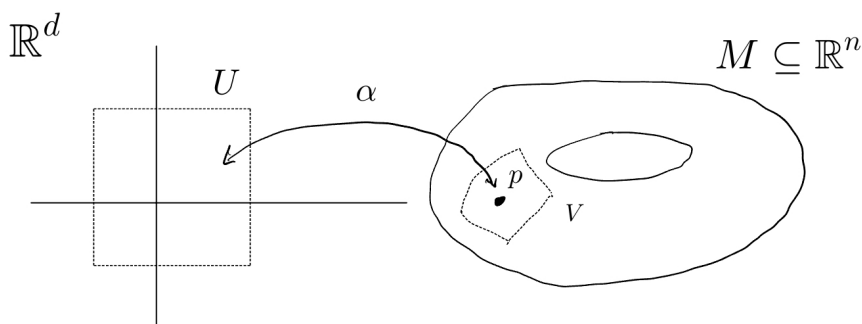


Figure 2.1: Visualisation of a Manifold and a coordinate patch

Example 2.1.2 (a function that is not a coordinate patch).

$M = \{(t^3, t^2) \mid t \in \mathbb{R}\}$, $\alpha : \mathbb{R} \rightarrow M$ with $\alpha(t) = (t^3, t^2)$. α is a smooth homeomorphism with $D\alpha(t) = (3t^2, 2t)$. However, $D\alpha(0) = (0, 0)$ has rank 0, hence α is not a coordinate

patch. Note that just because this function isn't a coordinate patch, does not mean M is not a manifold. In this case M is not a manifold but that must be shown separately (by considering open sets around $(0,0)$).

Example 2.1.3 (another function that is not a coordinate patch).

$\alpha(t) = (t^3, |t^3|)$ and $M = \alpha(\mathbb{R})$. α is a homeomorphism and of class C^2 yet for the same reason as the previous example, it is not a coordinate patch.

Example 2.1.4 (last non-example of a coordinate patch).

$\alpha(x, y) = (x^3, x^2, y)$, $M = \alpha(\mathbb{R}^2)$.

$$D\alpha(x, y) = \begin{bmatrix} 3x^2 & 0 \\ 2x & 0 \\ 0 & 1 \end{bmatrix}$$

If $x=0$, the derivative has rank 1, while if $x \neq 0$ it has rank 2. Since the derivative must have the same rank for each point, this is not a coordinate patch. If we consider $M \setminus \{(0, 0, t)\}$, then α is a coordinate patch.

Sometimes only a single $\alpha : U \rightarrow M$ suffices for the whole of M , in this case we call (M, α) a parameterized manifold. For example $M = \{(t, t^2) \in \mathbb{R}^2 \mid t \in \mathbb{R}\}$ with coordinate patch $\alpha(t) = (t, t^2)$

"Everyone knows what a curve is, until he has studied enough mathematics to become confused through the countless number of possible exceptions"
- Felix Klein

From now on we only consider smooth Manifolds.

2.2 An Equivalent Definition of a Manifold

For still they knew, and ought to have still remember'd The high Injunction not to taste that Fruit, Whoever tempted; which they not obeying, Incurr'd, what could they less, the penaltie, And manifold in sin, deserv'd to fall.

-John Milton (Paradise Lost) (Manifold also means "many and various")

Theorem 2.2.1. For $M \subset \mathbb{R}^n$, $d > 0$, $p \in M$ the following are equivalent:

1. $\forall p \in M \exists V \in \mathcal{T}_{\mathbb{R}^n}$, $V \subset M$, $U \in \mathcal{T}_{\mathbb{R}^d}$ and a function $\alpha : U \rightarrow V$ where α is:
 - (a) Smooth/ C^r
 - (b) Is a bijection with a continuous inverse
 - (c) $D\alpha(x)$ has rank $d \forall x \in U$
2. $\exists V \in \mathcal{T}_{\mathbb{R}^n}$ such that $p \in V$, and $\exists U \subset \mathbb{R}^n$,
 $\beta : U \rightarrow V$ is a diffeomorphism and $\beta(U \cap \mathbb{R}^d \times \{0\}) = M \cap \beta(U) = M \cap V$

Proof:

1) $\implies$ 2): Without loss of generality assume $\alpha(0) = p$ (We can translate the whole space to make this the case). $D\alpha(0)$ has rank $d \iff$ it has d linearly independent columns. $D\alpha \in \text{Mat}_{n \times d} \implies \exists B \in \text{Mat}_{n \times n-d}$ such that $[D\alpha(0), B] \in \text{Mat}_{n \times n}$ is invertible. Define the following map:

$$\beta : U \times \mathbb{R}^{n-d} \rightarrow \mathbb{R}^n$$

$$\beta(x, y) = \alpha(x) + By$$

The derivative of a linear transformation is itself, hence $D\beta(0) = [D\alpha(0), B]$ is invertible. By the inverse function theorem, $\exists \tilde{U}, \tilde{V} \subset \mathbb{R}^n$ such that $0 \in \tilde{U}$ and $\beta : \tilde{U} \rightarrow \tilde{V}$ is a diffeomorphism. $\tilde{U} \cap \mathbb{R}^d \times \{0\} \subset U \implies \beta : (\tilde{U} \cap \mathbb{R}^d \times \{0\}) = \alpha(\tilde{U} \cap \mathbb{R}^d \times \{0\})$ is an open neighbourhood of p and equals $M \cap \tilde{V}$ for some $\tilde{V} \in \mathcal{T}_{\mathbb{R}^n}$. Replace $\tilde{V}$ with $\tilde{V} \cap \tilde{V}$ and $\tilde{U}$ with $\beta^{-1}(\tilde{V}) \implies \beta(\tilde{U} \cap \mathbb{R}^d \times \{0\}) = \tilde{V} \cap M$

2) $\implies$ 1): Define $\tilde{U} = U \cap \mathbb{R}^d \times \{0\}$

$\alpha = \beta|_{\tilde{U}} \rightarrow \alpha(\tilde{U})$. Since α is a restriction of a homeomorphism, it is a homeomorphism. $\implies \beta(U) \cap M$ is open. $\alpha(x_1, x_2, \dots, x_d) = \beta(x_1, x_2, \dots, x_d, 0, \dots, 0)$. By the chain rule:

$$D\alpha = D\beta \circ \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 \\ 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 \end{bmatrix}$$

Clearly the above matrix will have rank d since it is the first d columns of an invertible matrix.

Definition 2.2.2 (Smooth functions). For $M \subset \mathbb{R}^n$, $N \subset \mathbb{R}^m$ $f : M \rightarrow N$ is smooth if $\exists V \in \mathcal{T}_{\mathbb{R}^n}, M \subset V, \tilde{f} : V \rightarrow \mathbb{R}^m$ is smooth such that $\tilde{f}|_M = f$

f is a diffeomorphism if it is smooth and so is its inverse. (Note that we do not require $\tilde{f} \circ \tilde{f}^{-1} = \text{id}$) only that $\tilde{f} \circ \tilde{f}^{-1}|_M = \text{id}$

Proposition 2.2.3. The following useful properties hold:

- $f : M \rightarrow N$, $A \subset M$ is a diffeomorphism $\implies f|_A : A \rightarrow f(A)$ is a diffeomorphism.
- Composition of smooth functions are smooth
- Composition of diffeomorphisms are diffeomorphisms

Corollary 2.2.4. Let $M \subset \mathbb{R}^n$, $d > 0$.

M is a manifold of dimension $d \iff \forall p \in M, \exists V \subset M, V \in \mathcal{T}_M, p \in V, U \in \mathcal{T}_{\mathbb{R}^d}$ and a diffeomorphism $\alpha : V \rightarrow U$

Proof:

" $\implies$ " By the previous theorem, we find $\mathbf{p} \in V \subset \mathbb{R}^n$ and $U \subset \mathbb{R}^n$ such that $\beta : U \rightarrow V$ is a diffeomorphism. By the above proposition, $\beta|_{U \cap \mathbb{R}^d \times \{0\}}$ is a diffeomorphism onto $V \cap M = \beta(U) \cap M$

" $\Longleftarrow$ " $\alpha : U \rightarrow V$ is a diffeomorphism $\implies$ it is a smooth homeomorphism

$\alpha^{-1} : V \rightarrow U$ is smooth $\iff \exists \tilde{\alpha}^{-1} : \tilde{V} \rightarrow \mathbb{R}^n$ is smooth ($\tilde{\alpha}$ is an extension of α).

$\tilde{\alpha}^{-1} \circ \alpha = \alpha^{-1} \circ \alpha = \text{id}|_U$

$\implies D\tilde{\alpha}^{-1}(\alpha(x)) \circ D\alpha(x) = I_d$. Since $D\alpha$ is a $d \times d$ matrix, hence it must have rank d by this calculation.

Proposition 2.2.5. *If $\exists$ a diffeomorphism f between open sets $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^n$, then $n = m$.*

Remark 2.2.6.

Note that in the above corollary technically would should say that the embedding of U into $\mathbb{R}^d$ is open in $\mathbb{R}^d$. If we stopped every time to consider whether we mean embeddings and things of the sort we would get bogged down very quickly. Usually it's clear from the context what we mean.

Proposition 2.2.7. *Let M be a manifold with boundary. Let $\alpha_i : V_i \rightarrow U_i$ $i=1,2$ be coordinate patches of M . Then $\alpha_2^{-1} \circ \alpha_1 : \alpha_1^{-1}(U_1 \cap U_2) \rightarrow \alpha_2^{-1}(U_1 \cap U_2)$ is a diffeomorphism. This can be viewed as a coordinate change.*

Proof: α_1 is a diffeomorphism $\implies \alpha_1|_{\alpha_1^{-1}(U_1 \cap U_2)} : \alpha_1^{-1}(U_1 \cap U_2) \rightarrow U_1 \cap U_2$ is a diffeomorphism since it is the restriction of a diffeomorphism. An identical argument applies to α_2 . Then $\alpha_2^{-1} \circ \alpha_1$ is a diffeomorphism as it is the composition of diffeomorphisms.

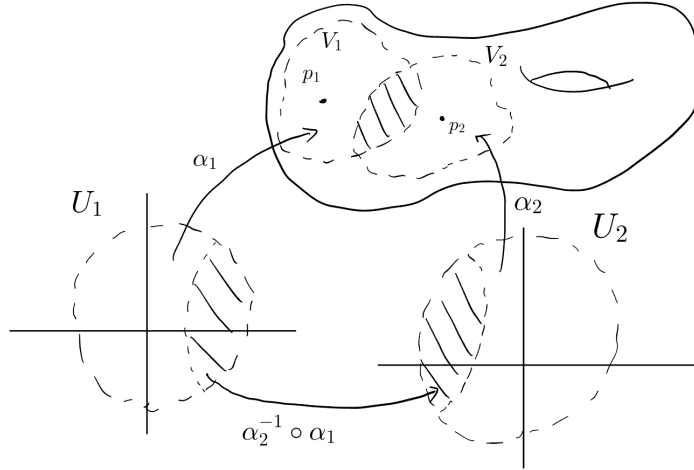


Figure 2.2: Visualisation of transition functions

2.3 Partitions of Unity

Partitions of unity can be thought of as smooth functions that assign a weight to each coordinate patch on a set, depending on how many coordinate patches are defined on that set. It's akin to interpolation. They allow us to "glue" together local data to get a global picture. Don't get too caught up with the proof of their existence. It's not massively insightful from our perspective. It's important to know when they exist and what their properties are.

Theorem 2.3.1 (Existence of Partitions of Unity). *Let $V \subset \mathbb{R}^n$ where $V = \bigcup_{\alpha \in A} U_\alpha$ with $U_\alpha \in \mathcal{T}_{\mathbb{R}^n}$. Then there exists $\varphi_1, \varphi_2, \dots$ with $\varphi_i : V \rightarrow [0, 1]$ with each having the following properties:*

1. For each $i \in \mathbb{N}, \exists \alpha \in A$ such that $S_i := \text{supp}(\varphi_i)$ (where "supp" is the support of φ_i) = the closure of $\{x \in V \mid \varphi_i(x) > 0\} \subset U_\alpha$
2. Each $p \in A$ has a neighbourhood intersecting finitely many S_i 's
3. $\sum_{i=1}^{\infty} \varphi_i(x) = 1 \forall x \in V$
4. S_i is compact $\forall i$
5. φ_i is smooth $\forall i$

$\{\varphi_i\}$ is called a partition of unity subordinate to $\{U_\alpha\}$.

Remark 2.3.2. By 2) the sum in 3) is locally finite i.e each point has a neighbourhood containing only finitely many S_i .

Lemma 2.3.3. With $\{U_\alpha\}$ as above, then $\exists p_1, p_2, \dots \in \mathbb{R}^n, \varepsilon_1, \varepsilon_2, \dots \in \mathbb{R}$ such that the following holds:

1. $\bigcup_{i=1}^{\infty} B(p_i, \varepsilon_i) = V$
2. Each $B(p_i, 2\varepsilon_i)$ is contained in a U_α
3. Each point $p \in V$ has neighbourhood intersecting finitely many $B(p_i, 2\varepsilon_i)$

Lemma 2.3.4. We can find $K_1 \subset K_2 \subset \dots \subset V$ such that K_i are compact, $K_i \subset K_{i+1}^\circ$ and $\bigcup K_i = V$

Proof: $K_i = \overline{B(0, i)} \setminus \bigcup_{x \in V^c} B(x, \frac{1}{2^i})$

Proof of lemma 2.3.3: Set $K_0 = K_{-1} = \emptyset$. Cover $K_i \setminus K_{i-1}^\circ$ with $B(x, \varepsilon_x)$ such that $B(x, 2\varepsilon_x) \subset (K_{i-2})^c$, $B(x, 2\varepsilon_x)$ is contained in some U_α . By compactness, we can find finitely many x_j^i, ε_j^i such that $B(x_j^i, \varepsilon_j^i)$ covers $K_i \setminus K_{i-1}^\circ$

Claim: $\{(x_j^i, \varepsilon_j^i)\}$ has the desired properties.

1. $\bigcup K_i \setminus K_{i-1}^\circ = V$, $(\bigcup_{i,j} B(x_j^i, \varepsilon_j^i)) \supset \bigcup_i K_i \setminus K_{i-1}^\circ \supset \bigcup_i K_i \setminus K_{i-1} = \bigcup_i K_i = V$
2. By construction

3. $B(x_j^i, 2\varepsilon_j^i)$ does not intersection $K_{i-2} \supset K_{i-3} \supset \dots$
 $\implies K_{i+1}$ intersects finitely many $B(x_j^i, 2\varepsilon_j^i)$, $(B(x_j^i, \varepsilon_j^i))$ cannot intersect K_{i-2}
 $\implies$ for any $p \in A$, $\exists i$ such that $p \in K_i \subset K_{i+1}^\circ$. K_{i+1}° is an open neighbourhood intersecting finitely many $B(x, 2\varepsilon)$'s. $\square$

Lemma 2.3.5. *Let $p \in \mathbb{R}^n$, $\varepsilon > 0$. Then $\exists \varphi : \mathbb{R}^n \rightarrow [0, 1]$ such that:*

- φ is smooth
- $\text{supp}(\varphi) \subset B(p, 2\varepsilon)$
- $\varphi > 0$ on $B(p, \varepsilon)$

Proof: Without loss of generality, let $p=0$, $\varepsilon = \frac{2}{3}$. Let φ be defined as:

$$\varphi(x) = \begin{cases} e^{\frac{1}{1-\|x\|^2}} & \|x\| < 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof of theorem 2.3.1: Find p_i, ε_i as in lemma 2.3.5 and take $\tilde{\varphi}_i$ as in lemma 2.3.4. We verify each condition except 3), we then modify $\tilde{\varphi}_i$.

1. $\text{supp}(\tilde{\varphi}_i) \subset B(p_i, 2\varepsilon_i) \subset U_\alpha$ for some α .
2. $\text{supp}(\tilde{\varphi}_i) = S_i \subset B(p_i, 2\varepsilon_i)$ and $B(p_i, 2\varepsilon_i)$ has the local finiteness property.
3. Below
4. $S_i \subset B(p_i, 2\varepsilon_i)$. Since S_i is closed and it's contained in a bounded set, it is closed and bounded and therefore compact.
5. Smooth by lemma 2.3.5.

To Prove 3) we first show $\psi(x) = \sum_{i=1}^{\infty} \tilde{\varphi}_i(x) > 0 \forall x \in V$. By lemma 2.3.3, $x \in B(p_i, \varepsilon_i)$ for some i and hence $\psi(x) \geq \tilde{\varphi}_i(x) > 0$. If we define $\varphi_i(x) := \frac{\tilde{\varphi}_i(x)}{\psi(x)}$ which still satisfies the 4 other conditions.

$$\sum_{i=1}^{\infty} \varphi_i(x) = \sum_{i=1}^{\infty} \frac{\tilde{\varphi}_i(x)}{\psi(x)} = \frac{1}{\psi(x)} \sum_{i=1}^{\infty} \tilde{\varphi}_i(x) = 1 \quad \square$$

Proposition 2.3.6 (Local Smoothness at Every Point Implies Global Smoothness). *Let $M \subset \mathbb{R}^n$ with $f : M \rightarrow \mathbb{R}^m$. Suppose f is locally smooth i.e $\forall p \exists V \subset M$, $p \in V$, $V \in \mathcal{T}_{\mathbb{R}^n}$ such that $f|_V : V \rightarrow \mathbb{R}^m$ is smooth then f is smooth.*

Proof: For each $x \in M$ we find $x \in V_x \subset \mathbb{R}^n$ and $\tilde{f}_x : V_x \rightarrow \mathbb{R}^m$ smooth such that $\tilde{f}|_{V_x \cap M} = f|_{V_x \cap M}$. Let $\{\varphi_i\}$ be a partition of unity subordinate to $\{V_x\}_{x \in M}$. For each i , $\exists V_{x_i}$ such that $\text{supp}(\varphi_i) \subset V_{x_i}$. Define

$$h_i(x) = \begin{cases} \varphi_i(x) \tilde{f}_{x_i}(x) & x \in U_{x_i} \\ 0 & \text{Otherwise} \end{cases}$$

$h_i : \bigcup V_x \rightarrow \mathbb{R}^m$ is smooth. Let $\tilde{f} = \sum_{i=1}^{\infty} h_i(x)$. For each $x \in \bigcup V_x$, $\exists$ a neighbourhood U_x intersecting finitely many $\text{supp}(\varphi_i) = S_i$. Let these be $S_1, \dots, S_N \implies \tilde{f}|_{U_x} = \sum_{i=1}^N h_i(x)$, hence $\tilde{f}$ is smooth. Moreover, for $x \in U_x \cap M$

$$\tilde{f}(x) = \sum_{i=1}^N \varphi_i(x) \tilde{f}_{x_i}(x) = \sum_{i=1}^N \varphi_i(x) f(x) = f(x) \left(\sum_{i=1}^N \varphi_i(x) \right) = f(x)$$

$$\bigcup V_x \supset M \implies \tilde{f}|_M = f \quad \square$$

Definition 2.3.7 (Convex Property). *P is a convex property if $S_1, \dots, S_k, \dots$ satisfy property P implies $\sum \lambda_i S_i$ satisfies P (where $\sum \lambda_i = 1$ and $\lambda_i \geq 0$)*

Remark 2.3.8. *Being an extension of f is a convex property, using partitions of unity allows us to globalise any convex property from local data.*

2.4 Manifolds with Boundary

Definition 2.4.1. *The following definitions are important for consider manifolds with boundary.*

- $\mathbb{H}^d := \mathbb{R}^{d-1} \times \mathbb{R}_{\geq 0} = \{(x_1, \dots, x_d) \mid x_d \geq 0\}$ (The Upper Half-Space)
- $\partial \mathbb{H}^d := \mathbb{R}^{d-1} \times \{0\} \subset \mathbb{H}^d$ (The Boundary of $\mathbb{H}^d$)
- $(\mathbb{H}^d)^\circ = \text{int}(\mathbb{H}^d) := \mathbb{R}^{d-1} \times \mathbb{R}_{>0} = \mathbb{H}^d \setminus \partial \mathbb{H}^d$

Definition 2.4.2 (d-Manifold with Boundary). *A subset $M \subset \mathbb{R}^n$ is a d-manifold with boundary if it is locally diffeomorphic to an open subset of $\mathbb{H}^d$, that is, $\forall p \in M, \exists V \in \mathcal{T}_M, U \in \mathcal{T}_{\mathbb{H}^d}$, with $p \in V, \cdot$. There also exists a diffeomorphism $\alpha : U \rightarrow V$.*

Note that by the above definition, a manifold without boundary is also a manifold with Boundary.

"If I had a world of my own, everything would be nonsense. Nothing would be what it is, because everything would be what it isn't. And contrary wise, what is, it wouldn't be. And what it wouldn't be, it would. You see?"

-Lewis Carroll (Alice's Adventures in Wonderland)

Proposition 2.4.3. *The condition above is equivalent to α being a smooth homeomorphism with derivative of rank d for all x in U.*

Lemma 2.4.4. *If U is an open subset of $\mathbb{H}^d$ and $\alpha : U \rightarrow \mathbb{R}^n$ is smooth, let $\tilde{\alpha} : \tilde{U} \rightarrow \mathbb{R}^n$ be a smooth extension. Then $D\tilde{\alpha}(x), x \in U$ does not depend on the extension (We can just call it $D\alpha(x)$).*

Proof: It is clear for $x \in \text{int}(\mathbb{H}^d)$. For $x \in \partial \mathbb{H}^d \cap U$, we write $\tilde{\alpha}(x) = (\tilde{\alpha}_1(x), \dots, \tilde{\alpha}_n(x))$.

$$(D\tilde{\alpha})_{i,j} = \frac{\partial \tilde{\alpha}_j}{\partial x_i}(x) = \lim_{\varepsilon \rightarrow 0} \frac{\tilde{\alpha}_j(x + \varepsilon \hat{e}_i) - \tilde{\alpha}_j(x)}{\varepsilon}$$

If $i \neq d$, the above does not depend on $\tilde{\alpha}$.

If $i=d$:

$$\frac{\partial \tilde{\alpha}_j}{\partial x_i}(x) = \lim_{\varepsilon \rightarrow 0^+} \frac{\tilde{\alpha}_j(x + \varepsilon e_d) - \tilde{\alpha}_j(x)}{\varepsilon}$$

In the above, the quotient that we're taking the limit of is not dependent on our extension.

□

Proof of Proposition 2.4.3: " $\implies$ " Let $\tilde{\alpha}$ and $\tilde{\alpha}^{-1}$ be smooth extensions of α and α^{-1} . Then $\tilde{\alpha}^{-1} \circ \tilde{\alpha}(x) = x$, $\forall x \in U \subset \mathbb{H}^d$ i.e $\tilde{\alpha}^{-1} \circ \tilde{\alpha} = f$ where f is some smooth function such that $f(x)=x$, $\forall x \in U$.

$D\tilde{\alpha}^{-1}(\tilde{\alpha}(x)) \circ D\tilde{\alpha}(x) = Df(x) = I_d \implies D\tilde{\alpha}(x) = D\alpha(x)$ has rank d .

" $\impliedby$ " An identical proof of the equivalent proposition for manifolds without boundary.

Definition 2.4.5. Let M be a manifold with boundary. p is an interior point if $\exists$ a coordinate patch $\alpha : U \rightarrow V$ such that $p = \alpha(x)$, $x \in \text{int}(\mathbb{H}^d)$. p is a boundary point if $\exists$ a coordinate patch $\alpha : U \rightarrow V$ such that $p = \alpha(x)$, $x \in \partial\mathbb{H}^d$

We define $M^\circ := \{x \in M \mid x \text{ is an interior point of } M\}$, $\partial M := \{x \in M \mid x \text{ is a boundary point of } M\}$
Note that these don't necessarily equal the topological boundary and interior of M in $\mathbb{R}^n$!

Example 2.4.6.

$M = \{(x, y) \in \mathbb{R}^2 \mid x = 0, y \in [0, 1]\} = \{0\} \times [0, 1]$ M is a manifold with boundary.

$\alpha_1 : [0, 1] \rightarrow \{0\} \times [0, 1]$, $\alpha_1(t) = (0, t)$

$\alpha_2 : [0, 1] \rightarrow \{0\} \times (0, 1]$, $\alpha_2(t) = (0, 1 - t)$

α_1, α_2 are coordinate patches covering M .

$\{0\} \times (0, 1) \subset M^\circ$

$\{0\} \times \{0, 1\} \subset \partial M$

With the following lemma, we find that the previous inclusions are actually equalities.

Lemma 2.4.7 ($M = M^\circ \cup \partial M$).

Proof: Clearly $M^\circ \cup \partial M \subset M$.

We wish to show that $M^\circ \cap \partial M = \emptyset$. Let $x \in M^\circ \cap \partial M \implies \exists$ coordinate patches $\alpha_1 : U_1 \rightarrow V_1$, $\alpha_2 : U_2 \rightarrow V_2$.

Let $x_1 \in U_1 \cap \text{int}(\mathbb{H}^d)$, $x_2 \in \partial\mathbb{H}^d$ such that $\alpha_1(x_1) = x = \alpha_2(x_2)$. This implies $\varphi = \alpha_2^{-1} \circ \alpha_1 : \alpha_1^{-1}(V_1 \cap V_2) \rightarrow \alpha_2^{-1}(V_1 \cap V_2)$ is a diffeomorphism with $\varphi(x_1) = x_2$.

φ is a diffeomorphism $\implies D\varphi(x_1)$ is invertible. By the inverse function theorem $\exists$ a neighbourhood $V \ni x_1$, $V \subset \text{int}(\mathbb{H}^d)$ such that $\varphi(V)$ is open in $\mathbb{R}^d$ but $\varphi(V) \subset \mathbb{H}^d$, $x_2 \in \partial\mathbb{H}^d \cap \varphi(V)$ which is a contradiction □

2.5 The Boundary of a Manifold is a Manifold

"And as for those parts that came from the earth, they shall return unto the earth again; and those that came from heaven, they also shall return unto those heavenly places. Whether it be a mere dissolution and unbinding of the manifold intricacies and entanglements of the

confused atoms; or some such dispersion of the simple and incorruptible elements..."

-Marcus Aurelius (Meditations) (Once again, by "manifold" he doesn't mean a set that's locally homeomorphic to euclidean space)

Theorem 2.5.1. *Let M be a d -manifold with boundary. Then ∂M is a $(d-1)$ -manifold without boundary.*

Proof: Let $p \in \partial M \subset M$. Let $\alpha : U \rightarrow V$ be a coordinate patch around p . Then $\alpha|_{U \cap \partial \mathbb{H}^d} : U \cap \partial \mathbb{H}^d \rightarrow \alpha(U \cap \partial \mathbb{H}^d) = \alpha(U) \cap \partial M = V \cap \partial M$ is a diffeomorphism. $\square$

Theorem 2.5.2 (Level Sets of Smooth Functions are Manifolds). *Let $U \in \mathcal{T}_{\mathbb{R}^n}$, $f : U \rightarrow \mathbb{R}$ be a smooth function. Let $M = \{x \in U \mid f(x) = 0\}$. Suppose $\forall x \in M, Df(x)$ has rank 1, then $N = \{x \in U \mid f(x) \geq 0\}$ is a manifold with boundary and $\partial N = M$, in particular, M is a manifold.*

Proof: Let $p \in N$. If $f(p) > 0$ then $f^{-1}((0, \infty)) \xrightarrow{\alpha=i^d} f^{-1}((0, \infty))$ is a coordinate patch around p .

If $f(p) = 0$, $Df(p) \neq 0 \implies \exists i$ such that $\frac{\partial f}{\partial x_i}(p) \neq 0$ (assume $i=n$). Define $F : U \rightarrow \mathbb{R}^n$, $F(x) = (x_1, \dots, x_{n-1}, f(x))$.

$$DF(p) = \begin{bmatrix} 1 & 0 & \cdots & \frac{\partial f}{\partial x_1}(p) \\ 0 & 1 & \cdots & \frac{\partial f}{\partial x_2}(p) \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \frac{\partial f}{\partial x_n}(p) \end{bmatrix} \text{ is invertible}$$

The inverse function theorem implies that $\exists U, V$ such that $F|_U : U \rightarrow V$ is a diffeomorphism. Note that $F|_U^{-1}(\mathbb{H}^d \cap V) = U \cap N$. Now set $\alpha = F|_U^{-1}(\mathbb{H}^d \cap V) \rightarrow U \cap N$. $\alpha(p_1, p_2, \dots, p_{n-1}, 0) = p$ and α is a diffeomorphism as it's a restriction of a diffeomorphism. α is our required coordinate patch. $\square$

Example 2.5.3. $f(x) = r^2 - \|x\|^2$. $\overline{B(0, r)} = \{x \in \mathbb{R}^n \mid f(x) \geq 0\}$.

$S^{n-1}(r) = \{x \in \mathbb{R}^n \mid f(x) = 0\}$. For $x \in S^{n-1}(r)$:

$Df(x) = (-2x_1, \dots, -2x_n) = -2(x_1, \dots, x_n) \neq 0$.

This implies $\overline{B(0, r)}$, $S^{n-1}(r)$ are manifolds with $\partial \overline{B(0, r)} = S^{n-1}(r)$

Example 2.5.4. $M = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y + e^z + \cos x = 0\}$.

The above is a manifold because if we define $F(x, y, z) = x^2 + y + e^z + \cos x$, its derivative is never 0, i.e M is a level set of a smooth function.

2.6 Tangent Spaces

Definition 2.6.1 (The Tangent Space of a Manifold). *Let $M \subset \mathbb{R}^n$ be a manifold with boundary. Let $p \in M$. $\alpha : U \rightarrow V$ is the chart around p with $\alpha(x_0) = p$. The tangent space of M at p is*

$$T_p M := \text{Im}(D\alpha(x_0)) \subset \mathbb{R}^n$$

Lemma 2.6.2. *This definition is not dependent on the coordinate patch chosen for p .*

"God ever geometrizes"

-Plato

Proposition 2.6.3. *Let M and N be manifolds with boundary. Let $f : M \rightarrow N$ be a smooth function. Then $Df(p) = D\tilde{f}(p)$ (where $\tilde{f}$ is an extension of f) defines a linear map between $T_p M$ and $T_{f(p)} N$ for all $p \in M$.*

Proof: Let $\alpha_1 : U_1 \rightarrow V_1$ and $\alpha_2 : U_2 \rightarrow V_2$ be coordinate patches around p and $f(p)$ respectively. Since $p \in V_1$, $f(p) \in f(V_1) \cap V_2$, we can "shrink" U_1 (and V_1), we can assume $f(V_1) \subset V_2$. $\alpha_2^{-1} \circ f \circ \alpha_1 : U_1 \rightarrow U_2$ is well defined and smooth.

Let $\tilde{f}$ be any extension of f , then $\alpha_2^{-1} \circ \tilde{f} \circ \alpha_1 = \alpha_2^{-1} \circ f \circ \alpha_1 = \varphi$. Then $\tilde{f} \circ \alpha_1 = \alpha_2 \circ \varphi = f \circ \alpha_1$ does not depend on the extension and we obtain

$$D\tilde{f}(p) \circ D\alpha_1(\alpha_1^{-1}(p)) = D\alpha_2(\alpha_2^{-1}(f(p))) \circ D\varphi(\alpha_1^{-1}(p)) = Df(p) \circ D\alpha_1(\alpha_1^{-1}(p))$$

$D\alpha_1(\alpha_1^{-1}(p))$ has image $T_p M$ and $D\alpha_2(\alpha_2^{-1}(f(p)))$ has image $T_{f(p)} N$

$\implies D\tilde{f}(p)$ restricted to $T_p M$ does not depend on $\tilde{f} \implies D\tilde{f}|_{T_p M} \subset T_{f(p)} N$. $\square$

Example 2.6.4. $M = \mathbb{R} \times \{0\} \subset \mathbb{R}^2$

$f : M \rightarrow \mathbb{R}$, $f(x) = 0$, $\tilde{f}_1(x, y) = y$, $\tilde{f}_2(x, y) = 0$

$\tilde{f}_1|_M = \tilde{f}_2|_M = f$ despite $D\tilde{f}_1 \neq D\tilde{f}_2$ but $T_p M = \text{span} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$

$D\tilde{f}_1|_{T_p M} = D\tilde{f}_2|_{T_p M} = 0$

Definition 2.6.5 (Tangent Bundle of M). *Let M be a manifold with boundary in $\mathbb{R}^n$. $TM = \{(x, v) \in M \times \mathbb{R}^n \mid v \in T_x M\}$*

Note that TM is the disjoint union (we don't care about the intersection of $T_{p_1} M$ and $T_{p_2} M$). We can define the canonical projection $\pi : TM \rightarrow M$ with $\pi^{-1}(\{x\}) = T_x M$

Example 2.6.6. $M = [0, 1]$, $TM = [0, 1] \times \mathbb{R}$

Example 2.6.7. $T_{(x,y)} \mathbb{S}^1 = \{(x, y, u, v) \in \mathbb{R}^4 \mid xu + yv = 0, x^2 + y^2 = 1\}$

2.7 Regular Values

Proposition 2.7.1 (Properties of TM).

1. TM is a manifold with boundary
2. $f : M \rightarrow N$ is smooth $\rightarrow f_* = Df : TM \rightarrow TN$ is smooth ($Df(p, v) = (f(p), Df(p) \cdot v)$)
3. $f : M \rightarrow N$, $g : N \rightarrow L$ are smooth $\implies D(g \circ f) = Dg \circ Df$

Definition 2.7.2 (Regular and Critical Values). *Let $f : M \rightarrow N$ be smooth. p is a regular value if $Df(x) : T_x M \rightarrow T_p N$ is surjective for all $x \in f^{-1}(p)$, otherwise p is a critical value.*

Theorem 2.7.3. $f : M \rightarrow N$ is smooth, M and N are a d_1 and a d_2 manifold without boundary, $p \in N$ is a regular value. Then $L = f^{-1}(\{p\})$ is a manifold, moreover $T_x L = \ker(Df(x) : T_x M \rightarrow T_p N)$

Proof: Let $x_0 \in L \subset M$ and $\alpha : U_1 \rightarrow V_1 \subset M$, $\beta : U_2 \rightarrow V_2 \subset N$ be charts around x_0 and p respectively, such that $f(V_1) \subset V_2$. Then $\beta^{-1} \circ f \circ \alpha = \varphi : U_1 \rightarrow U_2$ with $U_1 \subset \mathbb{R}^{d_1}$, $U_2 \subset \mathbb{R}^{d_2}$ is smooth.

p is regular value $\implies D\varphi(x_0)$ is surjective. We can therefore find a matrix B such that

$$\det \left(\begin{bmatrix} D\varphi(x_0) \\ B \end{bmatrix} \right) \neq 0$$

Define $\psi : U_1 \rightarrow U_2 \times \mathbb{R}^{d_1-d_2}$ by $\psi(x) = (\varphi(x), Bx)$. Then

$$D\psi(x_0) = \begin{bmatrix} D\varphi \\ B \end{bmatrix} \text{ by the inverse function theorem } \implies \psi^{-1} : \tilde{U}_2 \rightarrow \tilde{U}_1 \text{ is smooth}$$

Set $\alpha_L = \alpha \circ \psi^{-1}|_{\tilde{U}_2 \cap \{\beta^{-1}(p)\} \times \mathbb{R}^{d_1-d_2}}$. We claim that α_L is a chart for L around x_0 , $L \cap \alpha(\tilde{U}_1) = \alpha_L(\tilde{U}_2 \cap \{\beta^{-1}(p)\} \times \mathbb{R}^{d_1-d_2}) = \text{Im}(\alpha_L)$ and $\text{rank}(D\alpha_L) = d_1 - d_2$ (check this!)

Finally $f \circ \alpha_L(x) = p$, $\forall x \implies Df \circ D\alpha_L = 0 \implies \text{Im}(D\alpha_L) \subset \ker(Df)$.

$\dim(\ker(Df)) = d_1 - \dim(\text{Im}(Df))$ (by the rank-nullity theorem) $= d_1 - d_2 = \dim(\text{Im}(D\alpha_L)) \implies \text{Im}(D\alpha_L) = \ker(Df) \quad \square$

Remark 2.7.4.

The above proof strategy is common. The inverse function theorem tells us that to attain a chart, we have to define a function $\psi : U \rightarrow \mathbb{R}^d$ such that $\psi(0)$ is invertible. We "augment" some function f that we already have, with a function g such that $\psi = (f, g)$ defines a chart.

Theorem 2.7.5 (Sard's theorem). *Let $f : M \rightarrow N$ be smooth. Then the set of critical values of f has measure 0. The set of regular values is dense in N .*

Proof: Omitted, left as exercise (for masochists)

Chapter 3

Multilinear Algebra

3.1 Motivation for Stoke's Theorem

In studying manifolds, we hope to come to a generalisation of the following 3 theorems:

1. (Fundamental Theorem of Calculus)

$$\int_a^b f'(x)dx = f(b) - f(a)$$

2. (Gauss' Law)

$$\iiint \text{div}(\mathbf{E}) \cdot d\mathbf{V} = \iint \mathbf{E} \cdot d\mathbf{A}$$

3. (Green's Theorem)

$$\iint_D \left(\frac{\partial B}{\partial x} - \frac{\partial A}{\partial y} \right) dx dy = \oint_C A dx + B dy$$

All of the above can be expressed in the following form:

$$\int_M d\omega = \int_{\partial M} \omega$$

This known as the generalised Stoke's theorem (Or simply Stoke's theorem). Roughly speaking, in each of the above ω is some kind of volume element. Any volume element ought to satisfy the intuitive properties of a multilinear function as detailed below. We use the tools of linear algebra to aid us once again in developing our calculus on manifolds.

The further a mathematical theory is developed, the more harmoniously and uniformly does its construction proceed, and unsuspected relations are disclosed between hitherto separated branches of the science.

-David Hilbert

3.2 Multilinear Functions

Definition 3.2.1 (Multilinear Function). Let V be a d -dimensional vector space, a function $T : V \times V \times \dots \times V \rightarrow \mathbb{R} = T : V^k \rightarrow \mathbb{R}$ is called multilinear (or a k -tensor) if for every i , $T_i(\hat{v}_i) = (v_1, \dots, v_{i-1}, \hat{v}_i, \dots, v_k)$ is linear. $\mathcal{L}^k(V) = \{T : V^k \rightarrow \mathbb{R} \mid T \text{ is multilinear}\}$

Definition 3.2.2 (Dual Space of V). The dual space of V , denoted by V^* , is the set of all linear functions from V to $\mathbb{R}$.

Note that the above definition is equivalent to $V^* = \mathcal{L}^1(V)$

Proposition 3.2.3. $\mathcal{L}^k(V)$ is a vector space with $(f+g)(v_1, \dots, v_k) = f(v_1, \dots, v_k) + g(v_1, \dots, v_k)$ and $(\lambda f)(v_1, \dots, v_k) = \lambda f(v_1, \dots, v_k)$

Definition 3.2.4 (Dual Basis of V). Let $\{e_i\}$ be a basis for V . Define $e^j \in V^*$ by it being the row vector such that $e^j \left(\sum_{i=1}^k a_i e_i \right) = a_j$. More generally for $I = (i_1, \dots, i_k)$ define $e^I(v_1, \dots, v_k) = e^{i_1}(v_1) \cdot \dots \cdot e^{i_k}(v_k)$

Theorem 3.2.5. $\{e^I\}$, with $I \in \{1, \dots, d\}^k$ is a basis for $\mathcal{L}^k(V)$. $\dim(\mathcal{L}^k(V)) = d^k$

Proof:

1. Show e^I is multilinear
2. Show $v = \sum_{i=1}^d e^i(v) \cdot e_i$
3. Expand $T(v_1, \dots, v_k)$ using 2. and show the span of $\{e^I\} = \mathcal{L}^k(V)$
4. Evaluate $\sum \alpha_I e^I = 0$ at $e_{i_1}, \dots, e_{i_k}$ to demonstrate linear independence.

Definition 3.2.6 (Tensor Product). Let $f \in \mathcal{L}^k(V)$ and $g \in \mathcal{L}^\ell(V)$. We define $f \otimes g \in \mathcal{L}^{k+\ell}(V)$ by

$$(f \otimes g)(v_1, \dots, v_k, v_{k+1}, \dots, v_{k+\ell}) = f(v_1, \dots, v_k) \cdot g(v_{k+1}, \dots, v_{k+\ell})$$

Theorem 3.2.7. Let f, g and h be tensors.

1. $f \otimes (g \otimes h) = (f \otimes g) \otimes h$
2. $(\lambda f) \otimes g = \lambda(f \otimes g) = f \otimes (\lambda g)$
3. $(f + g) \otimes h = (f \otimes h) + (g \otimes h)$
4. $h \otimes (f + g) = (h \otimes f) + (h \otimes g)$
5. $e^I = e^{i_1} \otimes \dots \otimes e^{i_k}$

Remark 3.2.8. $\bigotimes_{k=0}^{\infty} \mathcal{L}^k(V)$ is an algebra freely generated by V^* .

Definition 3.2.9 (Pull-Back of an Operator). Let $A : V \rightarrow W$ be a linear map. We define the dual transformation $A^* : \mathcal{L}^k(W) \rightarrow \mathcal{L}^k(V)$ by $(A^*f)(v_1, \dots, v_k) = f(Av_1, \dots, Av_k)$ where $f \in \mathcal{L}^k(W)$

Theorem 3.2.10. Let A^* be as above, then:

1. A^* is linear.
2. $A^*(f \otimes g) = A^*f \otimes A^*g$
3. If $B : U \rightarrow V$ and $A : V \rightarrow W$ are linear, then $(AB)^* = B^*A^*$

3.3 Alternating Tensors

Definition 3.3.1. A tensor $f \in \mathcal{L}^k(V)$ is called:

- *symmetric* if $f(v_1, \dots, v_i, v_{i+1}, \dots, v_k) = f(v_1, \dots, v_{i+1}, v_i, \dots, v_k)$
- *alternating* if $f(v_1, \dots, v_i, v_{i+1}, \dots, v_k) = -f(v_1, \dots, v_{i+1}, v_i, \dots, v_k)$ (for adjacent entries)

Let $S^k V$ and $A^k V$ be the vector spaces of symmetric and alternating tensors respectively.

Definition 3.3.2 (The Symmetric Group). $S_k = \{\sigma : \{1, \dots, k\} \rightarrow \{1, \dots, k\} \mid \sigma \text{ a bijection}\}$.

A permutation $\sigma_i \in S_k$ is elementary if:

$$\sigma_i(\ell) = \begin{cases} i+1 & \text{if } \ell = i \\ i & \text{if } \ell = i+1 \\ \ell & \text{otherwise} \end{cases}$$

Lemma 3.3.3. Every $\sigma \in S_k$ is a composite of elementary permutations.

Theorem 3.3.4 (Existence of the Signum Function).

There exists a function $\text{sgn} : S_n \rightarrow \{\pm 1\}$ with the following properties:

1. $\text{sgn}(\sigma \circ \tau) = \text{sgn}(\sigma) \cdot \text{sgn}(\tau)$ i.e it is a group homomorphism.
2. $\text{sgn}(\sigma_i) = -1$
3. $\text{sgn}(\sigma) = \prod_i \text{sgn}(\sigma_i) = (-1)^m$ where σ is the product of m elementary permutations.
4. $\text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma)$
5. $\text{sgn}(\sigma) = -1$ if σ is a transposition.

Note that sgn is uniquely determined by conditions 1) and 2).

Algebraically, the sgn function is evaluated as follows:

$$\text{sgn}(\sigma) = \prod_{i < j} \text{sign}(\sigma(j) - \sigma(i)) = \prod_{i < j} \frac{\sigma(i) - \sigma(j)}{i - j}$$

Definition 3.3.5. For $f \in \mathcal{L}^k(V)$, $\sigma \in S_k$, $f^\sigma(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)})$

Lemma 3.3.6 ($\mathcal{L}^k(V)$ is a S_k -Representation (Compatible with the group structure of S_k)).

- $f^{\sigma \circ \tau} = (f^\tau)^\sigma$, $f^e = f$.
- $f \rightarrow f^\sigma$ is a linear map $\mathcal{L}^k(V) \rightarrow \mathcal{L}^k(V)$

Proposition 3.3.7.

$f \in \mathcal{L}^k(V)$ is symmetric $\iff f^\sigma = f \ \forall \sigma \in S_k$

$f \in \mathcal{L}^k(V)$ is alternating $\iff f^\sigma = \text{sgn}(\sigma) \cdot f \ \forall \sigma \in S_k$

Lemma 3.3.8. $f \in \mathcal{L}^k(V)$ is alternating $\iff f(v_1, \dots, v_k) = 0$ whenever $v_i = v_j$ for some $i \neq j$

Proof: " $\implies$ " Let σ be a permutation exchanging i and j and fixing everything else.
 $f^\sigma(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = f(v_1, \dots, v_j, \dots, v_i, \dots, v_k) = -f(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = f(v_1, \dots, v_i, \dots, v_j, \dots, v_k) \implies f(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = 0$
 " $\impliedby$ " For simplicity we take $k=2$. $0 = f(u + v, u + v) = f(u, u + v) + f(v, u + v) = f(u, u) + f(u, v) + f(v, u) + f(v, v) = f(u, v) + f(v, u) = 0 \quad \square$

Lemma 3.3.9. Let $\{e_i\}$ be a basis for V . Let $f \in A^k V$ and suppose $f(e_{i_1}, \dots, e_{i_k}) = 0$ for all $I = (i_1, \dots, i_k)$ with $i_1 < \dots < i_k$, then $f=0$.

Proof: It suffices to show that $f(e_{i_1}, \dots, e_{i_k}) = 0$ for any $I = (i_1, \dots, i_k)$. If there exists a repeated index, then it's 0 by the previous lemma, otherwise, there exists $\sigma \in S_k$ such that $i_{\sigma(1)} < \dots < i_{\sigma(k)}$. $\text{sgn}(\sigma)f(e_{i_1}, \dots, e_{i_k}) = f(e_{i_{\sigma(1)}}, \dots, e_{i_{\sigma(k)}}) = 0 \implies f(e_{i_1}, \dots, e_{i_k}) = 0$

Lemma 3.3.10. Let $I = (i_1, \dots, i_k)$ such that $i_1 < \dots < i_k$. $\psi_I = \sum_{\sigma \in S_k} \text{sgn}(\sigma)(e^I)^\sigma$ is the unique, alternating k -tensor satisfying:

$$\psi_I(e_{j_1}, \dots, e_{j_k}) = \begin{cases} 1 & \text{if } I = (j_1, \dots, j_k), j_1 < \dots < j_k \\ 0 & \text{otherwise} \end{cases}$$

for all $J = (j_1, \dots, j_k)$ with $j_1 < \dots < j_k$

Proof: ψ_I is a k -tensor as it's a sum of k -tensors.

To show that it's alternating:

$$\begin{aligned} (\psi_I)^\tau &= \sum_{\sigma \in S_k} \text{sgn}(\sigma) ((e^I)^\sigma)^\tau = \sum_{\sigma \in S_k} \text{sgn}(\tau) \text{sgn}(\tau\sigma) (e^I)^{\tau\sigma} \\ &= \text{sgn}(\tau) \sum_{\sigma \in S_k} \text{sgn}(\sigma) (e^I)^{\tau\sigma} = \text{sgn}(\tau) \psi_I \\ \psi_I(e_{j_1}, \dots, e_{j_k}) &= \sum_{\sigma \in S_k} \text{sgn}(\sigma) \underbrace{e^I(e_{j_{\sigma(1)}}, \dots, e_{j_{\sigma(k)}})}_{\begin{cases} 1 & \text{if } \sigma(i) = i \forall i \\ 0 & \text{otherwise} \end{cases}} \quad \square \end{aligned}$$

Remark 3.3.11. Note that in the above, because $\sigma \mapsto \tau\sigma$ is a bijection, the sum remains over all permutations.

Example 3.3.12.

$V = \mathbb{R}^2$ with basis e_1, e_2 .

$\psi(e_1, e_2) = f$

$f(ae_1 + be_2, ce_1 + de_2) = ad.f(e_1, e_2) + bc.f(e_2, e_1) = ad - bc$

Theorem 3.3.13. $\{\psi_I\}_{I \text{ is ascending}}$ form a basis of $A^k(V)$. In particular, $\dim(A^k(V)) = \binom{n}{k}$, where $n = \dim(V)$

Proof:

$\text{span}(\psi_I) = A^k$:

Let $f \in A^k(V)$ and set $f^I = f(e_{i_1}, \dots, e_{i_k})$. Then $f - \sum_{(I \text{ is ascending})} f^I \cdot \psi_I = 0$ by the previous 2 lemmas.

Linear Independence:

The equation above determines f^I uniquely. $\square$

Remark 3.3.14. If $k=n$, then $\dim(A^k(V))=1$ with basis $\psi^{(1,2,\dots,n)}$

Proposition 3.3.15. Let $B : V \rightarrow W$ be linear, then $B^* : \mathcal{L}^k(W) \rightarrow \mathcal{L}^k(V)$ restricts to $B^* : A^k(W) \rightarrow A^k(V)$ i.e B^*f is alternating if f is.

Theorem 3.3.16. Suppose $B : V \rightarrow V$, then $B^* : A^n(V) \rightarrow A^n(V)$ is multiplication by a real number. Then if $n=\dim(V)$, $B^*f = \det(B) \cdot f$, $f \in A^n(V)$

Proof: We only need to check for one $f \neq 0$, for instance $f = \psi^{(1,\dots,n)}$. Also it is enough to check $B^*\psi^I = \det \cdot f$ on $(e_1, \dots, e_n)$.

$$\begin{aligned} (B^*\psi^I)(e_1, \dots, e_n) &= \psi^I(Be_1, \dots, Be_n) \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) \underbrace{e^{(1,\dots,n)}(Be_{\sigma(1)}, \dots, Be_{\sigma(n)})}_{Be_{\sigma(1)} \cdot Be_{\sigma(2)} \cdot \dots \cdot Be_{\sigma(n)}} \end{aligned}$$

3.4 The Wedge Product

Theorem 3.4.1 (Existence of the Wedge Product). There exists an operation $A^k(V) \times A^\ell(V) \rightarrow A^{k+\ell}(V)$ with $(f, g) \mapsto f \wedge g$ satisfying:

1. $f \wedge (g \wedge h) = (f \wedge g) \wedge h$
2. $(f, g) \mapsto f \wedge g$ is multilinear if $\ell = k$
3. $f \wedge g = (-1)^{k\ell} g \wedge f$ where $f \in A^k(V)$ and $g \in A^\ell(V)$
4. If $\{e_i\}$ is a basis of V with dual basis $\{e^i\}$, then $\psi^I = e^{i_1} \wedge \dots \wedge e^{i_k}$ with $I = (i_1 < \dots < i_k)$

The wedge product is uniquely determined by conditions 1) and 4). Furthermore if $T : V \rightarrow W$ is linear, then $T^*(f \wedge g) = T^*f \wedge T^*g$

Proof: We define the averaging operator $\mathcal{A} : \mathcal{L}^k(V) \rightarrow \mathcal{L}^k(V)$ by $\mathcal{A}f = \sum_{\sigma \in S_k} \text{sgn}(\sigma) f^\sigma$. It satisfies:

- $\mathcal{A}$ is linear
- $\mathcal{A}f \in A^k(V)$
- If $f \in A^k(V)$, then $\mathcal{A}f = k!f$

We define the wedge product as:

$$f \wedge g := \frac{1}{k!\ell!} \mathcal{A}(f \otimes g), \text{ for } f \in \mathcal{A}^k(V), g \in \mathcal{A}^\ell(V)$$

We then verify that this has properties 1 – 4:

1) Proven below

2) By linearity of $\mathcal{A}$ and bi-linearity of $\otimes$

3) Let $\pi \in S_{k+\ell}$ such that $(\pi(1), \dots, \pi(k+\ell)) = (k+1, \dots, k+\ell, 1, 2, \dots, k)$, then $\text{sgn}(\pi) = (-1)^{k+\ell}$ and $(g \otimes f)^\pi = f \otimes g$

Thus

$$\begin{aligned} \mathcal{A}(g \otimes f) &= \sum_{\sigma \in S_{k+\ell}} \text{sgn}(\sigma) (g \otimes f)^\sigma = \sum_{\sigma \in S_{k+\ell}} \text{sgn}(\sigma\pi) (g \otimes f)^{\sigma\pi} \\ &= \text{sgn}(\pi) \sum_{\sigma \in S_{k+\ell}} \text{sgn}(\sigma) ((g \otimes f)^\pi)^\sigma = (-1)^{k+\ell} \mathcal{A}(f \otimes g) \end{aligned}$$

4) Also proven below

Lemma 3.4.2. *If $\mathcal{A}f = 0$, then $\mathcal{A}(f \otimes g) = 0$*

Proof: $\mathcal{A}(f \otimes g)(v_1, \dots, v_{k+\ell}) = \sum_{\sigma \in S_{k+\ell}} \text{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \cdot g(v_{\sigma(k+1)}, \dots, v_{\sigma(k+\ell)})$

$$= \sum_{i_1, \dots, i_\ell} \underbrace{\left[\sum_{\substack{\sigma \in S_{k+\ell} \\ (\sigma(k+1), \dots, \sigma(k+\ell)) = (i_1, \dots, i_\ell)}} \text{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \right]}_{\text{Choose one such } \sigma_0} g(v_{i_1}, \dots, v_{i_\ell})$$

$$\sum_{\tau \in S_k} \text{sgn}(\sigma_0 \tau) f(v_{\sigma_0(\tau(1))}, \dots, v_{\sigma_0(\tau(k))}) = \text{sgn}(\sigma_0) \mathcal{A}f(v_{\sigma_0(1)}, \dots, v_{\sigma_0(k)}) = 0 \quad \square$$

Corollary 3.4.3.

i) $\mathcal{A}(\mathcal{A}f \otimes g) = k! \mathcal{A}(f \otimes g)$

ii) $\mathcal{A}(f \otimes \mathcal{A}g) = \ell! \mathcal{A}(f \otimes g)$

Proof:

i) $\mathcal{A}(\mathcal{A}f - k!f) = 0 \implies \mathcal{A}((\mathcal{A}f - k!f) \otimes g) = 0$

ii) Use $\mathcal{A}(f \otimes g) = (-1)^{k\ell} \mathcal{A}(g \otimes f)$ and then apply i). $\square$

Proof of Theorem 3.4.1 part 1:

For $f \in \mathcal{A}^k(V)$, $g \in \mathcal{A}^\ell(V)$, $h \in \mathcal{A}^m(V)$

$$f \wedge (g \wedge h) = \frac{1}{k!(\ell+m)!} \mathcal{A}(f \otimes \frac{1}{(\ell)!(m)!} \mathcal{A}(g \otimes h))$$

$$= \frac{1}{k!\ell!m!} \mathcal{A}(f \otimes (g \otimes h)) = (f \wedge g) \wedge h$$

Proof of Theorem 3.4.1 part 4:

We claim that if $f_1, \dots, f_k \in V^*$ then $f_1 \wedge \dots \wedge f_k = \mathcal{A}(f_1 \otimes \dots \otimes f_k)$. We prove this claim by induction. It's evidently true for $k=1$.

We must show that it being true for $k=n$, implies it's true for $n=k+1$.

$$\begin{aligned} f_1 \wedge \dots \wedge f_k \wedge f_{k+1} &= \mathcal{A}(f_1 \otimes \dots \otimes f_k) \wedge f_{k+1} = \\ \frac{1}{k!} \mathcal{A}(\mathcal{A}(f_1 \otimes \dots \otimes f_k) \otimes f_{k+1}) &= \mathcal{A}(f_1 \otimes \dots \otimes f_{k+1}) \quad \square \end{aligned}$$

To show uniqueness:

By the bi-linearity of $\wedge$, it is determined by its image on a basis. It therefore suffices to show that $\psi^I \wedge \psi^J$ is determined by conditions 1-4.

By 4) $\psi^I \wedge \psi^J = e_1^i \wedge \dots \wedge e_k^i \wedge e_1^j \wedge \dots \wedge e_k^j$

By anti-commutativity, the right-hand side of the above is of the form $\pm e_1^{m_1} \wedge \dots \wedge e_{k+\ell}^{m_{k+\ell}}$ where $m_1 \leq m_2 \leq \dots \leq m_{k+\ell}$.

If $m_i = m_{i+1}$, then $e^{m_i} \wedge e^{m_i} = -e^{m_i} \wedge e^{m_i} = 0$ hence the right-hand side equals 0. Otherwise the right-hand side equals ψ^M .

Finally, let $T : V \rightarrow W$

$$\begin{aligned} k!\ell!T^*(f \wedge g) &= T * \mathcal{A}(f \otimes g) = T^* \sum_{\sigma} \text{sgn}(\sigma) (f \otimes g)^{\sigma} \\ &= \sum_{\sigma} \text{sgn}(\sigma) (T * (f \otimes g))^{\sigma} \\ &= \sum_{\sigma} \text{sgn}(\sigma) T * f \otimes T * g)^{\sigma} \\ &= \mathcal{A}(T^*f \otimes T^*g) = k!\ell!(T^*f \wedge T^*g) \quad \square \end{aligned}$$

Chapter 4

Differential Forms

4.1 Properties of Differential Forms

Let $M \subset \mathbb{R}^n$ be a smooth manifold with boundary.

Definition 4.1.1. *A differential form of order k is a smooth function*

$$\omega : \{(p, v_1, \dots, v_k) \in M \times \underbrace{\mathbb{R}^n \times \mathbb{R}^n \times \dots \times \mathbb{R}^n}_{k\text{-times}} \mid v_i \in T_p M\} \rightarrow \mathbb{R}$$

Such that $\forall p : \omega_p = \omega(p, v_1, \dots, v_k)$ is an alternating tensor $: (T_p M)^k \rightarrow \mathbb{R}$

$\Omega^k(M) = \{\omega \mid \omega \text{ is a smooth differential form of order } k\}$
 $\Omega_\delta^k(M)$ is the same as above except the forms aren't necessarily smooth.

Remark 4.1.2. $\Omega^0(M) = C^\infty(M)$

We can think of ω as $\omega : M \rightarrow \{A^k(T_p M)\}$

Lemma 4.1.3. $\Omega^k(M)$ is a vector space with point-wise addition and scalar multiplication.

Example 4.1.4. Let $\{e_i\}$ be the standard basis for $\mathbb{R}^n$ and let $(x^1, \dots, x^n) \in \mathbb{R}$ so that $e^i(x) = x^i$. Let $I = \{i_1, \dots, i_k\}$ be an ascending multi-index. $\omega(p, v_1, \dots, v_k) := \psi^I(v_1, \dots, v_k)$ is a differential k -form. We denote it by $dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_k} := \omega$

Proposition 4.1.5 (Pull-Back of a differential operator is well-defined). Let $f : M \rightarrow N$ be a smooth map. Define $f^* : \Omega^k(N) \rightarrow \Omega^k(M)$ as follows:

$$(f^*\omega)_p(v_1, \dots, v_k) := \omega_{f(p)}(Df(p)v_1, \dots, Df(p)v_k) \iff (f^*\omega)_p = (Df)^*\omega_{f(p)}$$

f^* is a well defined linear map.

Proof: $Df(p)v_i \in T_{f(p)}N$ hence the expression in the middle makes sense. Linearity is clear. It remains to show that $f^*\omega$ is smooth. $f^*\omega$ is a composite $T^{(k)}M \rightarrow T^{(k)}N \xrightarrow{\omega} \mathbb{R}$. It therefore suffices to show that the map $T^{(k)}M \rightarrow T^{(k)}N$ is smooth. Suppose $M \in \mathbb{R}^n$, $N \subset \mathbb{R}^m$. Let

$\tilde{f}$ be a smooth extension of f , then f^* extends to $\tilde{f}^*$.

$$\begin{aligned}\tilde{f}^* : \mathcal{U} \times (\mathbb{R}^n)^k &\rightarrow \mathcal{V} \times (\mathbb{R}^m)^k \\ (\mathbf{p}, \mathbf{v}_1, \dots, \mathbf{v}_k) &\rightarrow (\tilde{f}(\mathbf{p}), D\tilde{f}(\mathbf{p})\mathbf{v}_1, \dots, D\tilde{f}(\mathbf{p})\mathbf{v}_k)\end{aligned}$$

is the composition:

$$\mathcal{U} \times (\mathbb{R}^n)^k \xrightarrow{\tilde{f} \times D\tilde{f} \times \text{id}} \mathcal{V} \times \text{Lin}(\mathbb{R}^n, \mathbb{R}^m) \times (\mathbb{R}^m)^k \xrightarrow{\text{id} \times \text{ev}^k} \mathcal{V} \times (\mathbb{R}^m)^k$$

where ev^k is the evaluation map. Hence our map is smooth.

Proposition 4.1.6.

$$\cdot \text{id}_M^* = \text{id}_{\Omega^k(M)}$$

$$\cdot (f \cdot g)^* = g^* \cdot f^* \text{ for } M \xrightarrow{g} N \xrightarrow{f} L \text{ all smooth with } M, N, L \text{ manifolds with boundary.}$$

Proposition 4.1.7. *An element $\omega \in \Omega_\delta^k(M)$ is smooth if $\forall \mathbf{p} \in M, \exists$ a coordinate patch $\alpha : \mathcal{U} \rightarrow \mathcal{V} \subset M$ around p such that $\alpha^*\omega \in \Omega_\delta^k(\mathcal{U})$ is smooth.*

Proof: To show that ω is smooth, it suffices to show it is smooth at each $(\mathbf{p}, \mathbf{v}_1, \dots, \mathbf{v}_k) \in T^{(k)}M$.

Note that $T^{(k)}\mathcal{V} \subset T^{(k)}M$ is open, hence it suffices to show the restriction $\omega|_{\mathcal{V}} : T^{(k)}\mathcal{V} \rightarrow \mathbb{R}$ is smooth. $\omega|_{\mathcal{V}} = (\alpha^{-1})^* \underbrace{\alpha^*\omega|_{\mathcal{V}}}_{=\alpha^*\omega}$ and hence $\omega|_{\mathcal{V}}$ is smooth. $\square$

Proposition 4.1.8. *Let $\mathcal{U} \in \mathcal{T}_{\mathbb{R}^d}$, $\omega \in \Omega_\delta^k(\mathbb{R}^d)$*

For each p , $\omega_p \in A^k(\mathbb{R}^d)$ and hence $\omega_p = \sum_I \omega_{p,I} \cdot \psi^I$. Then ω is smooth if and only if $p \mapsto \omega_{p,I} \in \mathbb{R}$ are smooth for all ascending multi-indices I .

In previous notation $\omega(\mathbf{p}) = \sum_{I=(i_1 < \dots < i_k)} \omega_{p,I} \cdot \underbrace{dx^{i_1} \wedge \dots \wedge dx^{i_k}}_{\psi^I \in A^k(T_p \mathbb{R}^n)}$ or $\omega = \sum_I \omega_I \cdot dx^I$

Proposition 4.1.9. *Let $\omega \in \Omega^k(M), \eta \in \Omega^\ell(M)$, then $\omega \wedge \eta \in \Omega^{k+\ell}(M)$ where $(\omega \wedge \eta)_p(\mathbf{v}_1, \dots, \mathbf{v}_{k+\ell}) = (\omega_p \wedge \eta_p)(\mathbf{v}_1, \dots, \mathbf{v}_{k+\ell})$*

Proof: We need to show $\omega \wedge \eta$ is smooth. Recall it suffices to show that the restriction of $\alpha^*(\omega \wedge \eta)$ is smooth for some chart $\alpha : \mathcal{U} \rightarrow \mathcal{V}$, $\mathcal{U} \subset \mathbb{H}^d$, $\alpha^*(\omega) = \sum \omega_I dx^I$ and $\alpha^*(\eta) = \sum \eta_J dx^J$. $\alpha^*(\omega \wedge \eta) = \alpha^*(\omega) \wedge \alpha^*(\eta) = \sum_{I,J} \omega_I \eta_J \underbrace{dx^I \wedge dx^J}_{\pm dx^K \text{ for some } K}$ $\square$

Proposition 4.1.10. *If $f : M \rightarrow N$ is smooth, then $f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$*

Proof:

$$\begin{aligned}f^*(\omega \wedge \eta)(\mathbf{p}) &= Df(\mathbf{p})^*(\omega \wedge \eta)_{f(\mathbf{p})} \\ &= Df^*(\mathbf{p})(\omega_{f(\mathbf{p})} \wedge \eta_{f(\mathbf{p})}) \\ &= Df(\mathbf{p})^* \omega_{f(\mathbf{p})} \wedge Df(\mathbf{p})^* \eta_{f(\mathbf{p})} \\ &= f^*(\omega)_p \wedge (f^*\eta)_p \\ &= (f^*\omega \wedge f^*\eta)_p\end{aligned}$$

Remark 4.1.11. $\Omega^\cdot(M) = \bigoplus_{k=0}^{\infty} \Omega^k(M)$ admits an associative, anti-commutative algebra structure with $\omega \wedge \eta = (-1)^{k\ell} \eta \wedge \omega$

- $\Omega^0(M) = C^\infty$
- $f : M \rightarrow N \implies f^* : \Omega^\cdot(N) \rightarrow \Omega^\cdot(M)$ preserves all the above structure (It is an algebra homomorphism).
- $\Omega^k(M) = 0$ for $k > \dim(M)$

4.2 The Exterior Derivative of 0-Forms

Our goal is to define an "exterior derivative" $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ such that:

1. $df = Df$, $f \in C^\infty$
2. $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$
3. $d^2 = 0$

Example 4.2.1.

$M = \mathbb{R}^3$. $\omega = f \cdot dy \wedge dz$.

1. implies that

$$\begin{aligned} df = Df &\implies df(v) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \cdot \begin{pmatrix} v^1 \\ v^2 \\ v^3 \end{pmatrix} = \frac{\partial f}{\partial x} dx(v) + \frac{\partial f}{\partial y} dy(v) + \frac{\partial f}{\partial z} dz(v) \\ &\implies df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz \\ d\omega &= df \wedge dy \wedge dz + f \wedge \underbrace{d(dy \wedge dz)}_{d(dy) \wedge dz = 0} = \frac{\partial f}{\partial x} dx \wedge dy \wedge dz + \underbrace{\frac{\partial f}{\partial y} dy \wedge dy \wedge dz}_{=0} + \underbrace{\frac{\partial f}{\partial z} dz \wedge dy \wedge dz}_{=0} \\ &= \frac{\partial f}{\partial x} dx \wedge dy \wedge dz \end{aligned}$$

More formally, $f : M \rightarrow \mathbb{R}$, $Df_p : T_p M \rightarrow T_{f(p)} \mathbb{R} \ni (f(p), v)$.

Recall with $\{e_i\} = 1$ as a basis for $\mathbb{R}$, we defined $dx \in \Omega^1(\mathbb{R})$ by $dx(p, v) = v \cdot dx^1(v_1) = x^1(v_1)$, let us denote $dx = dx^1$

Definition 4.2.2. Let $f \in C^\infty(M)$, we define $df \in \Omega^1(M)$ by $df := f^* dx$

Unpacking the above definition we obtain that $df_p(v) = Df(p) \cdot v$ where $Df(p) : T_p M \rightarrow T_{f(p)} \mathbb{R} \simeq \mathbb{R}$ is a $1 \times n$ matrix.

Proposition 4.2.3. Let $M \subset \mathbb{U} \in \mathcal{T}_{\mathbb{H}^d}$, if $\{dx^1, \dots, dx^d\} \subset \Omega^1(M)$, then $df = \sum_{i=1}^d \frac{\partial f}{\partial x^i} \cdot dx^i$

Proof: $df(p)v = Df(p) \cdot v = \sum_{i=1}^d \frac{\partial f}{\partial x^i}(p) v^i = \sum_{i=1}^d \frac{\partial f}{\partial x^i}(p) dx^i(v)$

Proposition 4.2.4. Let $F : M \rightarrow N$ be a smooth map between smooth manifolds with boundary. Then $F^* df = dF^* f = d(f \circ F)$.

Proof: $F^*df = F^*(f^*dx) = (f \circ F)^*dx = d(f \circ F)$

Remark 4.2.5. $f : M \rightarrow \mathbb{R}$ then $f^*\varphi$ where $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ with $\varphi(x) = x \implies df = f^*d\varphi$

Lemma 4.2.6. Let $\pi^i : M \rightarrow \mathbb{R}$ be the projection function with $M \in \mathcal{T}_{\mathbb{R}^n}$. $\pi^i(x^1, \dots, x^i, \dots, x^n) = x^i$. Then $dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_k} = (d\pi^{i_1}) \wedge \dots \wedge (d\pi^{i_k}) \forall I = (i_1 < \dots < i_k)$

Proof:

$$\begin{aligned} (d\pi^{i_1} \wedge \dots \wedge d\pi^{i_k})_p(v_1, \dots, v_k) &= (d\pi_p^{i_1} \wedge \dots \wedge d\pi_p^{i_k})(v_1, \dots, v_k) \\ d\pi_p^i &= \underbrace{D\pi_{(p)}^i}_{(0, \dots, 1, \dots, 0)} v = v^i = e^i(v) \\ \implies (d\pi_p^{i_1} \wedge \dots \wedge d\pi_p^{i_k})(v_1, \dots, v_k) &= (e^{i_1} \wedge \dots \wedge e^{i_k})(v_1, \dots, v_k) \quad \square \end{aligned}$$

By convention, we denote by x^i the function $\pi^i : \mathbb{R}^n \rightarrow \mathbb{R}$ and likewise for the restriction to some open $U \subset \mathbb{R}^n$. In particular, any $\omega \in \Omega^k(U)$ is of the form:

$$\omega = \sum_{I \text{ ascending}} \omega_I dx^I, \quad \omega_I \in C^\infty(U)$$

Note that $dx^{i_1} \wedge \dots \wedge dx^{i_k}$ still makes sense even if I is not ascending.

Proposition 4.2.7. $d : C^\infty(M) \rightarrow \Omega^1(M)$ is linear.

Proof: Let $\lambda \in \mathbb{R}$ and define $F(x, y) = \lambda x + y$. Then $dF = \lambda dx + dy$. Let $f, g \in C^\infty$ with $(f \times g)(x) = (f(x), g(x))$. Then:

$$\begin{aligned} d(\lambda f + g) &= d(F \circ (f \times g)) = (f \times g)^*dF = (f \times g)^*(\lambda dx + dy) \\ &= (f \times g)^*dx + (f \times g)^*dy \\ &= \lambda(f \times g)^*(\pi^1)^*dx + dg \\ &= \lambda(\pi^1 \circ f \times g)^*dx + dg \\ &= \lambda f^*dx + dg \\ &= \lambda df + dg \end{aligned}$$

Proposition 4.2.8. $d(f \cdot g) = f \cdot dg + g \cdot df$

Proof: Define $F : \mathbb{R}^2 \rightarrow \mathbb{R}$, $F(x, y) = xy$. $dF = xdy + ydx$

$$\begin{aligned} d(f \cdot g) &= d(F \circ (f \times g)) = (f \times g)^*dF = (f \times g)^*(x \cdot dy + y \cdot dx) \\ &= (f \times g)^*x \cdot (f \times g)^*dy + (f \times g)^*y \cdot (f \times g)^*dx \\ &= f \cdot dg + g \cdot df \quad \square \end{aligned}$$

4.3 The Exterior Derivative

Theorem 4.3.1. *Let M be a manifold with boundary. There exists a unique linear transformation $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ defined for all $k \geq 0$ such that:*

1. *If $f \in \Omega^0(M) = C^\infty(M)$, then $df_p(v) = Df(p)v$ ($df = f^*dx$)*
2. *If $\omega \in \Omega^k(M), \eta \in \Omega^\ell$ then $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ (The Leibniz Rule)*
3. $d(d\omega) = 0$

Moreover, if $F : M \rightarrow N$ is any smooth map between smooth manifolds, then $d(F^\omega) = F^*(d\omega)$, $\omega \in \Omega^k(N)$ (Naturality)*

Proof: We first prove the special case where $M \subset \mathbb{U} \in \mathcal{T}_{\mathbb{H}^d}$.

To prove uniqueness, let $\omega \in \Omega^k(M)$,

$$\begin{aligned} \omega &= \sum_I \omega_I dx^I \\ d\omega &= \sum_I (d(\omega_I dx^I)) \\ &= \sum_I d\omega_I \wedge dx^I + \omega_I d(dx^I) \\ \text{But } d(dx^I) &= d(dx^{i_1} \wedge \dots \wedge dx^{i_k}) = \sum_\ell \pm dx^{i_1} \wedge \dots \wedge \underbrace{d(dx^{i_\ell})}_{=0} \wedge \dots \wedge dx^{i_k} \\ d\omega &= \sum_I d\omega_I \wedge dx^I = \sum_\ell \sum_{i_1 < \dots < i_k} \frac{\partial \omega_I}{\partial x^\ell} dx^\ell \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k} \end{aligned}$$

We then prove existence:

$$\omega = \sum_I \omega_I dx^I \text{ and define } d\omega := \sum_I d\omega_I \wedge dx^I$$

We can now prove statements 1-3.

1. ω is a 0 form (by convention $dx^\emptyset = 1$)
2. Both sides of 2 are linear in ω and linear in η . It suffices to consider $\omega = \omega_I dx^I$, $\eta = \eta_J dx^J$. ("The sum is just along for the ride"- Florian)
There are 2 cases: $I \cap J = \emptyset$ and $\exists \sigma$ which brings $I \cup J$ into the ascending index M and this $dx^I \wedge dx^J = \text{sgn}(\sigma) dx^M$.

$$\begin{aligned} \omega \wedge \eta &= \omega_I \eta_J \cdot \text{sgn}(\sigma) dx^M \\ d(\omega \wedge \eta) &= d(\omega_I \eta_J \cdot \text{sgn}(\sigma)) dx^M \\ &= \text{sgn}(\sigma) \omega_I d\eta_J \wedge dx^M + \text{sgn}(\sigma) \eta_J d\omega_I \wedge dx^M \\ &= \omega_I d\eta_J \wedge dx^I \wedge dx^J + \eta_J d\omega_I \wedge dx^I \wedge dx^J \\ &= (-1)^k \omega_I dx^I \wedge d\eta_J dx^J + (d\omega_I \wedge dx^J) \wedge (\eta_J dx^I) \\ &= (-1)^k \omega \wedge d\eta + d\omega \wedge \eta \end{aligned}$$

In the case that $I \cap J \neq \emptyset$, $dx^I \wedge dx^J = 0$ and the same proof gives $0=0$.

3. By linearity, it suffices to consider $\omega = \omega_I dx^I$.

$$\begin{aligned}
d(d(\omega_I dx^I)) &= d(d\omega_I \wedge dx^I) \\
&= d(d\omega_I) \wedge dx^I + \underbrace{d\omega_I d(1 \cdot dx^I)}_{=0} \\
d(d\omega_I) &= d\left(\sum_{i=1}^d \frac{\partial \omega_I}{\partial x^i} dx^i\right) \\
&= \sum_{i=1}^d d\left(\frac{\partial \omega_I}{\partial x^i}\right) dx^i = \sum_{i=1}^d \sum_{j=1}^d \left(\frac{\partial^2 \omega_I}{\partial x^j \partial x^i}\right) dx^j \wedge dx^i \\
&= \sum_{i>j} \left(\frac{\partial^2 \omega_I}{\partial x^j \partial x^i} - \frac{\partial^2 \omega_I}{\partial x^i \partial x^j}\right) dx^j \wedge dx^i = 0 \quad \square
\end{aligned}$$

To prove naturality let $F : U \rightarrow V$ be smooth with $U \subset \mathbb{R}^{d_1}, V \subset \mathbb{R}^{d_2}$, $\omega \in \Omega^k(M)$.

We proceed by induction on k : $k=0$: $d(F^*f) = F^*df$ (by definition).

To show k being true implies $k+1$ is true we can use linearity to take $\omega = \eta \wedge dx^i$.

$$d(F^*\omega) = d(F^*(\eta \wedge dx^i)) = d(F^*\eta \wedge F^*dx^i)$$

$$= d(F^*\eta) \wedge F^*dx^i + (-1)^k F^*\eta \wedge \underbrace{d(F^*dx^i)}_{=0}$$

$$F^*(d\omega) = F^*(d\eta \wedge dx^i) + (-1)^k \eta \wedge d(dx^i) = F^*(d\eta \wedge dx^i) = F^*(d\eta) \wedge F^*(dx^i)$$

For the case of M in general:

To show existence Let $\alpha : U \rightarrow V$ be a chart around $p \in M$. Define $(d\omega)_p := ((\alpha^{-1})^* d(\alpha^* \omega))_p$.

We claim that this is well-defined (independent of choice of chart).

Proof: Suppose we have α, β both charts around p . By restricting their codomains, we can assume that $\beta = \alpha \circ \varphi$ with $\varphi : U_1 \rightarrow U_2$ a diffeomorphism. We then have that:

$$\begin{aligned}
(\beta^{-1})^* d(\beta^* \omega) &= (\varphi^{-1} \circ \alpha^{-1})^* d(\alpha \circ \varphi)^* \omega \\
&= (\alpha^{-1})^* ((\varphi^{-1})^* d(\varphi^* \circ (\alpha^* \omega))) \\
&= (\alpha^{-1})^* \underbrace{(\varphi^{-1})^* \varphi^*}_{=id} d(\alpha^* \omega) \\
&= (\alpha^{-1})^* d(\alpha^* \omega)
\end{aligned}$$

We claim that d satisfies 1-3. The proof uses that α^* commutes with all operations.

For naturality for general M , the above definition and independence of choice of chart implies that $d\alpha^* \omega = \alpha^* d\omega$ whenever α is a chart.

For general M , it suffices to compare $dF^* \omega = F^* d\omega$ in a chart $\alpha : U \rightarrow M$ i.e

$$\begin{aligned}
\alpha^* dF^* \omega &= \alpha^* F^* d\omega \\
&= d(F \circ \alpha)^* \omega = (F \circ \alpha)^* d\omega
\end{aligned}$$

By restriction, we can assume $F \circ \alpha = \beta \circ \varphi$ where $\beta : V \rightarrow N$ is a chart and $\varphi : U \rightarrow V$ is smooth.

$$\begin{aligned} d(\beta \circ \varphi)^* \omega &= (\beta \circ \varphi)^* d\omega \\ &= d\varphi^* \beta^* \omega = \varphi^* \beta^* d\omega \\ &= \varphi^* d\beta^* \omega = \varphi^* d\beta^* \omega \end{aligned}$$

The proof for the uniqueness of M is omitted. Requires proof of the lemma that $\omega \in \Omega^k(M)$, $M \in \mathbb{R}^N$, $\omega = \sum \omega_I dx^{i_1} \wedge \dots \wedge dx^{i_k}$ non-uniquely for ω_I smooth.

Example 4.3.2.

$M = \{(x, y, z) \mid (e^x x)^2 + y^2 + z^2 = 1\}$ How can we write down differential forms on this "thing"?

$M \xrightarrow{x, y, z} \mathbb{R}$, $\omega = y \cdot dy \in \Omega^1(M)$. $\eta = y \cdot dy \in \Omega^1(\mathbb{R}^3)$. In the case of ω dy should more properly be $d(y|_M) = d(\iota^* y) \implies \omega = (\iota^* y)(d\iota^* y) = \iota^*(y dy) = \iota^* \eta$. Very often we shall simply write y instead of $y|_M$. Then $y \cdot dy, dx, dz \in \Omega^1(M)$. Are there relations? Informally, we take the relation that should hold on the manifold and apply the exterior derivative to it.

$$0 = d((e^x x)^2 + y^2 + z^2 - 1) = 2e^x x(e^x x dx + e^x dx) + 2y dy + 2z dz$$

More formally we define $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ with $F(x, y, z) = (e^x x)^2 + y^2 + z^2 - 1$. This function has the property that $F|_M = 0 = \iota^* F = 0 \implies d(\iota^* F) = \iota^* dF = 0$.

In practical terms:

$$\Omega^1(M) = \{f_x \cdot dx + f_y \cdot dy + f_z \cdot dz\} \Big/ (\text{the relations given by the structure of the manifold.})$$

$$\dot{\Omega}(M) = \{f_x dy \wedge dz + f_y dx \wedge dz + f_z dx \wedge dy\} \Big/ (\text{the relations given by the structure of the manifold.})$$

Chapter 5

Vector Calculus

5.1 Vector Fields

Definition 5.1.1 (Vector Field). *A vector field is a smooth function $X : M \rightarrow TM$ such that $X(p) \in T_p M$, $\mathfrak{X} = \{X : M \rightarrow TM \mid X \text{ is a vector field}\}$*

Remark 5.1.2. *We have a natural map $\pi : TM \rightarrow M$. A vector field is a smooth section of π . That is, a smooth map $X : M \rightarrow TM$ such that $\pi \circ X = \text{id}_M$*

Example 5.1.3.

Let M open in $\mathbb{R}^d$, $TM = M \times \mathbb{R}^d$, $X \in \mathfrak{X}(M)$, $X(p) = (p, v_p)$. We abuse notation to say that $X : p \mapsto v_p$, $X = (x^1, \dots, x^d)$

For the rest of this section, let M be open in $\mathbb{R}^d$:

$$\Omega^0(M) = C^\infty(M)$$

$$\Omega^1(M) = \left\{ \sum_{i=1}^d f_i dx^i \mid f_i \in C^\infty(M) \right\}$$

$\vdots$

$$\Omega^{n-1}(M) = \left\{ \sum_{i=1}^d f_i dx^1 \wedge \dots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \dots \wedge dx^d \mid f_i \in C^\infty \right\}$$

$$\Omega^n(M) = \left\{ u dx^1 \wedge \dots \wedge dx^d \mid u \in C^\infty(M) \right\}$$

We can now define the following isomorphisms for $M \subset \mathbb{R}^n$

$$h_1 : \mathfrak{X}(M) \rightarrow \Omega^1(M)$$

$$X = (x^1, \dots, x^d) \mapsto \sum x^i dx^i$$

$$h_{n-1} : \mathfrak{X}(M) \rightarrow \Omega^{n-1}(M)$$

$$X \mapsto \sum (-1)^{i-1} x^i dx^1 \wedge \dots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \dots \wedge dx^d$$

$$h_n : C^\infty(M) \rightarrow \Omega^n(M)$$

$$u \mapsto u dx^1 \wedge \dots \wedge x^n$$

Note that these isomorphisms are not necessarily natural i.e they don't necessarily commute with pullback.

5.2 Operations on Vectors fields

Recall from vector calculus that:

- $\text{grad} : C^\infty(M) \rightarrow \mathfrak{X}(M)$
 $f \mapsto \left(\frac{\partial f}{\partial x^1}, \dots, \frac{\partial f}{\partial x^d} \right) = \nabla f$

- $\text{div} : \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$
 $X \mapsto \nabla \cdot X$

- for $M \subset \mathbb{R}^3$
 $\text{curl} : \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$
 $X \mapsto \nabla \times X$

Theorem 5.2.1. *The following commutativity diagrams commute:*

$$\begin{array}{ccc} C^\infty(M) & \xrightarrow{\text{id}} & \Omega^0(M) \\ \text{grad} \downarrow & & \downarrow d \\ \mathfrak{X}(M) & \xrightarrow{h_1} & \Omega^1(M) \end{array}$$

$$\begin{array}{ccc} \mathfrak{X}(M) & \xrightarrow{h_{n-1}} & \Omega^{n-1} \\ \text{div} \downarrow & & \downarrow d \\ \mathfrak{X}(M) & \xrightarrow{h_2} & \Omega^n(M) \end{array}$$

$$\begin{array}{ccc}
\mathfrak{X}(M) & \xrightarrow{h_1} & \Omega^1(M) \\
\text{curl} \downarrow & & \downarrow d \\
\mathfrak{X}(M) & \xrightarrow{h_2} & \Omega^2(M)
\end{array}$$

Proof:

$$\cdot h_1(\text{grad}(f)) = \sum \frac{\partial f}{\partial x^i} dx^i = df$$

$$\begin{aligned}
\cdot d(h_{n-1}(X)) &= d \left(\sum_i (-1)^{i-1} x^i dx^1 \wedge \dots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \dots \wedge dx^d \right) \\
&= \sum_i \sum_j (-1)^{i-1} \frac{\partial x^i}{\partial x^j} dx^j \wedge dx^1 \wedge \dots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \dots \wedge dx^d \\
&= \sum_i (-1)^i \frac{\partial x^i}{\partial x^i} dx^i \wedge dx^1 \wedge \dots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \dots \wedge dx^d \\
&= \left(\sum_i \frac{\partial x^i}{\partial x^i} \right) dx^1 \wedge \dots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \dots \wedge dx^d \\
&= \text{div}(X) dx^1 \wedge \dots \wedge dx^d = h_n(\text{div}(X))
\end{aligned}$$

$$\begin{aligned}
\cdot d(h_1(X)) &= d(x^1 dx^1 \wedge x^2 dx^2 \wedge x^3 dx^3) \\
&= \frac{\partial x^1}{\partial x^2} dx^2 \wedge dx^1 + \frac{\partial x^1}{\partial x^3} dx^3 \wedge dx^1 + \dots \\
&= \left(\frac{\partial x^3}{\partial x^2} - \frac{\partial x^2}{\partial x^3} \right) dx^2 \wedge dx^3 + \dots \\
&= h_2(\text{curl}(X))
\end{aligned}$$

Corollary 5.2.2. $\text{curl}(\text{grad}(f)) = 0$, $\text{div}(\text{curl}(X)) = 0$

Proof: Equivalent to $d^2 = 0$

5.3 Integrating Forms

Our goal for this section is to define what we mean by $\int_M \omega$ for $\omega \in \Omega^d(M)$ where M is a d -manifold with boundary.

Our idea is to suppose that M is compact. We can then cover M with finitely many coordinate patches $\alpha_i : U_i \rightarrow V_i$. We choose a partition of unity $\{\varphi_i\}$ subordinate to $\{V_i\}$. Then:

$$\int_M \omega = \int_M \sum_i \varphi_i \cdot \omega = \sum_i \int_{V_i} \varphi_i \omega, \quad \varphi_i \omega = 0 \text{ outside of } V_i$$

$$= \sum_i \int_{U_i} \underbrace{\alpha_i^*(\varphi_i \cdot \omega)}_{=u_i dx^1 \wedge \dots \wedge dx^d} = \sum_i \int_{\mathbb{R}^d} u_i dx^1 \wedge \dots \wedge dx^n$$

We must ask whether this procedure is independent of the choices we made along the way. The answer is "almost".

Example 5.3.1.

$M = \mathbb{R}$, $\omega = f(x)dx$ where f is some positive function. Our chart is $\alpha_+(x) = x$

$$\alpha^* \omega = f(x).dx \implies \int \omega = \int_{-\infty}^{\infty} f(x)dx > 0$$

If we instead choose $\alpha_-(x) = -x$ as our chart we get the following:

$$\alpha^* \omega = \alpha^* f \alpha^* dx = f(\alpha(x))d(\alpha^* x) = f(-x)d(-x) = -f(-x)dx < 0, \implies \int \omega = \int_{-\infty}^{\infty} -f(-x)dx < 0$$

Clearly we need some sense of orientation.

Recall the following important properties from integration on $\mathbb{R}^d$

Theorem 5.3.2. $\exists$ an idea of integration such that $\int_{\mathbb{R}^d} \dots dx^1 \dots dx^n \mapsto \mathbb{R}$ satisfying

- Fubini's Theorem:

$$\int_{\mathbb{R}} f(x^1, \dots, x^n).dx = \int_{\mathbb{R}^{d-\ell}} \left(\int_{\mathbb{R}^{\ell}} f(x^1, \dots, x^n) dx^1 \dots dx^{\ell} \right) dx^{\ell+1} \dots dx^n$$

- The Fundamental Theorem of Calculus:

$$\int_{\mathbb{R}} f'(x).dx = f(\infty) - f(-\infty)$$

- The Change of Variables Theorem: Let $g : U_1 \rightarrow U_2$ be a diffeomorphism such that $\text{supp}(f) \subset U_2$, then:

$$\int_{\mathbb{R}^n} f(g(x))|\det(Dg)|dx^1 \dots dx^n = \int_{\mathbb{R}^n} f(x)dx^1 \dots dx^n$$

Definition 5.3.3. We say that $\omega \in \Omega^k(M)$ has compact support if $\text{supp } \omega = \overline{\{p \in M \mid \omega_p \neq 0\}}$ is compact. Let $\Omega_c^k(M)$ denote the space of k -forms with compact support.

Definition 5.3.4. Let $M \in \mathcal{T}_{\mathbb{H}^d}$, $\omega \in \Omega_c^d(M)$, we define:

$$\int_M \omega = \int_{\mathbb{R}^n} u dx^1 \dots dx^n, \text{ where } \omega = u \cdot dx^1 \wedge \dots \wedge dx^n$$

Chapter 6

Orientated Manifolds

6.1 Orientation-Preserving Diffeomorphisms

Definition 6.1.1 (Orientation-Preserving Diffeomorphisms). Let $F : M \rightarrow N$ be a diffeomorphism between open sets $M, N \subset \mathbb{H}^d$. We call F orientation preserving if $\det(DF(p)) > 0 \forall p \in M$. It's orientation reversing if $\det(DF(p)) < 0$

Remark 6.1.2. $\det(DF(p))$ is continuous, hence it has a constant sign on connected components.

Proposition 6.1.3. Let F be as above and orientation preserving. Then

$$\int_M F^* \omega = \int_N \omega \text{ for all } \omega \in \Omega_c^d(N)$$

Proof: with $\omega = u dx^1 \wedge \dots \wedge dx^d$

$$(F^* \omega)_p = (F^* u)_p DF(p)^* dx^1 \wedge \dots \wedge dx^d = \int_{\mathbb{R}^d} (u \circ F)(\det(DF(p))) dx^1 \wedge \dots \wedge dx^d$$

. Then

$$\int_M F^* \omega = \int_{\mathbb{R}^d} (u \circ F)(\det(DF)) dx^1 \wedge \dots \wedge dx^d, \int_N \omega = \int_{\mathbb{R}^d} u dx^1 \wedge \dots \wedge dx^d$$

They are equal if $\det(DF) = |\det(DF)|$. $\square$

Remark 6.1.4. If F is orientation reversing, then $\int_M F^* \omega = - \int_N \omega$

6.2 Orientable Manifolds

Let M be a manifold without boundary.

Definition 6.2.1 (Orientable Manifold). Let $\alpha_i : U_i \rightarrow V_i$ be two coordinate patches. We say that α_1, α_2 overlap positively if:

$$\alpha_2^{-1} \circ \alpha_1 : \alpha_1^{-1}(V_1 \cap V_2) \rightarrow \alpha_2^{-1}(V_1 \cap V_2), \text{ is orientation preserving}$$

If M can be covered by coordinate patches that pairwise overlap positively we say that M is orientable. The choice of such a covering is called an orientation.

Remark 6.2.2. An orientable manifold without boundary is denoted by $(M, \{\alpha_i\})$

Lemma 6.2.3. Let $(M, \{\alpha_i\})$ be an orientated manifold with boundary. Say a patch $\beta : U \rightarrow V$ is positive if it overlaps positively with all α_i . Then $\{\beta \mid \beta \text{ a positive patch}\}$ is an orientation containing $\{\alpha_i\}$ and not properly contained in any other orientation (i.e maximal)

Definition 6.2.4. An orientation on a 0-dimensional manifold is the choice of a function $\sigma : M \rightarrow \{\pm 1\}$

Example 6.2.5.

($d=1$), $M = [0, 1]$ with patches $\alpha_1 : [0, 1] \rightarrow M$ with $\alpha_1(t) = t$, $\alpha_2 : [0, 1] \rightarrow M$ with $\alpha_2(t) = 1 - t$. $(\alpha_2^{-1} \circ \alpha_1)(t) = -t$.

If we allow open sets in $(-\infty, 0]$ in definition of charts and orientation for 1-manifold.

Definition 6.2.6 (Amended Definition of Orientable Manifold). A 1-manifold with boundary is orientable if we can find a cover with positively overlapping patches whose domains are open sets in $[0, \infty)$ or $(-\infty, 0]$

Theorem 6.2.7. All 1-manifolds are orientable

Proof: Omitted for now. Uses the fact that all 1-manifolds are diffeomorphic to disjoint unions of $S^1, \mathbb{R}, [0, 1]$ and $[0, 1]$

Definition 6.2.8. For orientable $M \subset \mathbb{R}^d$, the single chart $\alpha = \text{id} : M \rightarrow M$ gives M its standard orientation.

Remark 6.2.9. There exist non-orientable manifolds

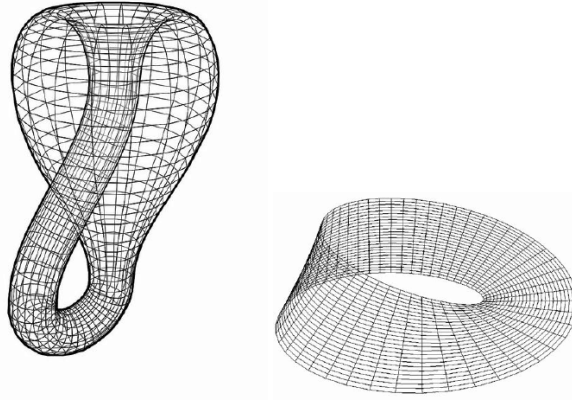


Figure 6.1: The Klein Bottle (left) and the Mobius Strip (right) are non-orientable manifolds

Set $\tau : \mathbb{H}^d \rightarrow \mathbb{H}^d$, $\tau(x^1, \dots, x^d) = (-x^1, \dots, x^d)$. For $d=1$, $\tau : [0, \infty) = (-\infty, 0]$

Given a coordinate patch α , we obtain another coordinate patch $\alpha \circ \tau$

Lemma 6.2.10. Let $(M, \{\alpha_i\})$ be an orientated manifold with boundary. Then $\{\alpha_i \circ \tau\}$ is also an orientation different from $\{\alpha_i\}$. We usually denote this by $(M, \{\alpha_i \circ \tau\}) = -M = -(M, \{\alpha_i\})$

Remark 6.2.11. For 0-manifolds we can change σ to $-\sigma$

6.3 A Technical Lemma

Lemma 6.3.1. *Let $(M, \{\alpha_i\})$ be an orientated manifold and $\alpha : U \rightarrow V$ a coordinate patch where U is connected. Suppose $\exists \alpha_i : U_i \rightarrow V_i$ and $x \in U$ such that $\alpha(x) \in V_i$ and $\det(D(\alpha_i^{-1} \circ \alpha)(x)) > 0$. Then $\{\alpha_i\} \cup \{\alpha\}$ is an orientation i.e α overlaps positively with all α_i .*

Proof: Define $\lambda : U \rightarrow \{\pm 1\}$ as follows. For $y \in U$, we let $\lambda(y)$ be the sign of the $\det D(\alpha_j \circ \alpha)(y)$ for any positive patch $\alpha_j : U_j \rightarrow V_j$ such that $\alpha(y) = p \in V_j$. This is well-defined since for any other positive patch α_k we obtain:

$$\det D(\alpha_j^{-1} \circ \alpha)(y) = \det D(\alpha_j^{-1} \circ \alpha_k \circ \alpha_k^{-1} \circ \alpha) = \underbrace{\det D(\alpha_j^{-1} \circ \alpha_k)(p)}_{>0 \text{ since } \alpha_j, \alpha_k \text{ positively overlap}} \cdot \det D(\alpha_k^{-1} \circ \alpha)(y)$$

It is clear that λ is continuous. Since it is locally constant and U is connected λ is constant.

Corollary 6.3.2. *Let M be orientated, $\alpha : U \rightarrow V$ a coordinate patch with U connected. Then either α or $\alpha \circ \tau$ is positive.*

Proof: By the previous lemma, being positive only depends on the sign of $\det D(\alpha_i^{-1} \circ \alpha)$ at a single point. If it is not less than 0, then $\det D(\alpha^{-1} \circ \alpha \circ \tau) = \det D(\alpha^{-1} \circ \alpha) \cdot \underbrace{\det D\tau}_{<0}$ is.

6.4 Orientations Extend to the Boundary of a Manifold

Proposition 6.4.1. *Let M be a manifold with boundary. Suppose $\text{int}(M) = M \setminus \partial M$ is orientable, then so is M . More precisely, if $\mathcal{A} = \{\alpha_i\}$ is an orientation on $\text{int}(M)$, then there exists a unique orientation $\mathcal{B}$ of M such that $\mathcal{A} \subset \mathcal{B}$*

Proof: Define $\mathcal{B} := \{\beta : U \rightarrow V \mid \beta \text{ is a coordinate patch on } M \text{ positively overlapping with every } \alpha_i\}$. " $\mathcal{A} \subset \mathcal{B}$ " left as exercise.

" $\mathcal{B}$ is an orientation": Note that if $\beta : U \rightarrow V$ is a patch on M , then $\beta|_{U \cap (\mathbb{H}^d)^\circ}$ is a patch on $\text{int}(M)$. β overlaps positively with α_i if and only if $\beta|_{U \cap (\mathbb{H}^d)^\circ}$ does.

Let $\beta_1, \beta_2 \in \mathcal{B}$, $\beta_i : U_i \rightarrow V_i$. We have to show that they overlap positively. Without loss of generality $V_1 = V_2 = V$. We need $\Phi = \beta_1^{-1} \circ \beta_2 : U_2 \rightarrow U_1$ is orientation preserving.

$$\Phi|_{U_2 \cap (\mathbb{H}^d)^\circ} = (\beta_1|_{U_1 \cap (\mathbb{H}^d)^\circ})^{-1} \circ \beta_2|_{U_2 \cap (\mathbb{H}^d)^\circ}$$

hence it is orientation preserving. This implies that $D\Phi : U_2 \rightarrow \mathbb{R}$ is continuous.. $D\Phi(U_2) \subset \mathbb{R} \setminus \{0\}$ and $D\Phi(U_2 \setminus \partial \mathbb{H}^d) \subset \mathbb{R}^+ \implies D\Phi(U_2) = D\Phi(\overline{U_2 \setminus \partial \mathbb{H}^d}) \subset \overline{D\Phi(U_2 \setminus \mathbb{H}^d)} \subset \mathbb{R}^+$.

Finally we must show that β s cover M . Let $p \in M$ and pick a chart $\alpha : U \rightarrow V$ around p . Without loss of generality assume U is connected. Then either β or $\beta \circ \tau$ overlaps positively with some and hence any α_i . $\square$

Corollary 6.4.2. *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be smooth and 0 a regular value. Then $f^{-1}([0, \infty)) = M$ carries a natural orientation.*

Remark 6.4.3.

With $M = \{x \mid f(x) \geq 0\}$ and $\alpha : U \rightarrow V \subset M$ a chart. α is positive $\iff \det(D\alpha(x)) > 0 \forall x \in U \cap \text{int}(\mathbb{H}^d)$

Proposition 6.4.4. *Let $(M, \{\alpha_i\})$ be a manifold with orientation and boundary. Then $(\partial M, \alpha_i|_{\partial\mathbb{H}^d \cap U_i})$ is an orientated manifold.*

Proof: Let $\mathcal{A} = \{\alpha_i|_{\partial\mathbb{H}^d \cap U_i}\}$. We want to show that $\mathcal{A}$ is an orientation. $\mathcal{A}$ covers ∂M , since $\{\alpha_i\}$ covers M . It remains to show that $\alpha_i|_{\partial\mathbb{H}^d \cap U_i}$ overlap positively. Define $\varphi := \alpha_1^{-1} \circ \alpha_2 : U \rightarrow U'$ with $U = \alpha_2^{-1}(V_1 \cap V_2)$ and $U' = \alpha_1^{-1}(V_1 \cap V_2)$. $\det(D\varphi_{U \cap \partial\mathbb{H}^d}) > 0$ since $\varphi_{U \cap \partial\mathbb{H}^d}$ is the transition function between $\alpha_1|_{U_1 \cap \partial\mathbb{H}^d}$ and $\alpha_2|_{U_2 \cap \partial\mathbb{H}^d}$ and $\varphi_{U \cap \partial\mathbb{H}^d} : U \cap \partial\mathbb{H}^d \rightarrow U' \cap \partial\mathbb{H}^d$. We know $\det(D\varphi) > 0$. Writing $\varphi = (\varphi_1, \dots, \varphi_d)$, then $\varphi_d(x^1, \dots, x^{d-1}, 0) = 0$ and $\varphi_d(x^1, \dots, x^{d-1}, x^d) \geq 0 \forall x^d \geq 0$. At $x = (x^1, x^2, \dots, x^{d-1}, 0)$, we have that:

$$D\varphi_d(x) = \left(0, 0, \dots, \underbrace{\frac{\partial \varphi_d}{\partial x^d}}_{\geq 0}\right)$$

$$D\varphi(x) = \begin{pmatrix} \frac{\partial \varphi_1}{\partial x^1} & \dots & \frac{\partial \varphi_1}{\partial x^{d-1}} & * \\ \vdots & \ddots & \vdots & * \\ \frac{\partial \varphi_{d-1}}{\partial x^1} & \dots & \frac{\partial \varphi_{d-1}}{\partial x^{d-1}} & * \\ 0 & \dots & 0 & \frac{\partial \varphi_d}{\partial x^d} \end{pmatrix}$$

The $(d-1) \times (d-1)$ matrix in the top left has positive determinant, and since the bottom right element is positive, $\det D\varphi_{U_1 \cap \partial\mathbb{H}^d} > 0$

Definition 6.4.5 (Induced Orientation on the Boundary).

The above proof constructs orientation on ∂M by restriction. The induced orientation on ∂M is $(-1)^d$ times the restricted one.

Definition 6.4.6 (Orientation Preserving Diffeomorphism).

Let $f : M \rightarrow N$ be a diffeomorphism between orientable manifolds with boundary $(M, \{\alpha_i\}), (N, \beta_j)$. We say f is orientation preserving if $\{f \circ \alpha_i\}$ are positive charts with respect to the orientation $\{\beta_j\}$

Example 6.4.7.

$D^d = \{x \in \mathbb{R}^d \mid \|x\|^2 \leq 1\}$ is orientable.

Orientation on $M \iff$ a partition of bases of $T_p M$ into positive and negative ones. $\iff$ a function $\{(0, \text{basis of } T_p M) \rightarrow \{\pm 1\}\} \iff$ a nowhere vanishing d -form.

6.5 Volume Forms

Definition 6.5.1 (Volume Form).

Let M be a d -manifold, $\omega \in \Omega^d(M)$ such that $\omega_p \neq 0 \forall p \in M$. Then ω is called a volume form.

Proposition 6.5.2. *Any volume form induces an orientation defining $\alpha : U \rightarrow V$ to be positive if $\alpha^*\omega = u dx^1 \wedge \dots \wedge dx^d$ where $u(x) > 0 \forall x \in U$*

Remark 6.5.3.

This is equivalent to demanding that $\int \alpha^*(f\omega) \geq 0 \forall f : M \rightarrow \mathbb{R}^+$

Proof of Proposition: Let α_1, α_2 be 2 positive patches. Without loss of generality let $V_1 = V_2$. We need to show that $\Phi = \alpha_1^{-1} \circ \alpha_2$ is orientation preserving. We know that

$$\alpha_1^*\omega = u_1 dx^1 \wedge \dots \wedge dx^d \quad (u_1 > 0)$$

$$\alpha_2^*\omega = u_2 dx^1 \wedge \dots \wedge dx^d \quad (u_2 > 0)$$

Applying $(\alpha_1^{-1})^*$ to $\alpha_1^*\omega$ we get

$$\omega|_V = (\alpha_1^{-1})^*(u_1 dx^1 \wedge \dots \wedge dx^d)$$

$$\begin{aligned} \underbrace{u_2}_{>0} dx^1 \wedge \dots \wedge dx^d &= \alpha_2^*((\alpha_1^{-1})^*(u_1 dx^1 \wedge \dots \wedge dx^d)) \\ &= (\alpha_1^{-1} \circ \alpha_2)^*(u_1 dx^1 \wedge \dots \wedge dx^d) \\ &= \Phi^*(u_1 dx^1 \wedge \dots \wedge dx^d) \\ &= \underbrace{\Phi^*u_1}_{>0} (\det D\Phi) dx^1 \wedge \dots \wedge dx^d \\ &\implies \det D\Phi > 0 \end{aligned}$$

To see why positive charts cover, let $p \in M, \alpha : U \rightarrow V$ be a patch around p with U connected. Then $\alpha^*\omega = u dx^1 \wedge \dots \wedge dx^d$ with either $u > 0$ or $u < 0$. In the latter case we consider $\alpha \circ \tau$. $\square$

Proposition 6.5.4. *Any orientation is induced by a volume form.*

Proof: Let $\{\alpha_i : U_i \rightarrow V_i\}$ be an orientation. We pick a partition of unity $\{\varphi_i\}$ subordinate to $\{V_i\}$. After reindexing we can assume that $\text{supp } \varphi_i \subset V_i$

$$\omega = \sum_i \underbrace{\varphi_i \cdot (\alpha_i^{-1})^*(dx^1 \wedge \dots \wedge dx^d)}_{\text{this has its support in } V_i \text{ so it can be smoothly extended by 0}}$$

ω is smooth clearly. We then show that it induces $\{\alpha_i\}$. Let $\beta : U \rightarrow V$ be positive with respect to $\{\alpha_i\}$. Then

$$\beta^*\omega = \sum_i \underbrace{\beta^*\varphi_i}_{>0} \cdot \underbrace{\beta^*((\alpha_i^{-1})^* dx^1 \wedge \dots \wedge dx^d)}_{=(\alpha_i^{-1} \circ \beta)^* dx^1 \wedge \dots \wedge dx^d}$$

$(\alpha_i^{-1} \circ \beta)^* > 0$ since $\alpha_i^{-1} \circ \beta$ is a transition function whose derivative has a positive determinant

$$\beta^* \omega = \underbrace{\left(\sum_i \beta^* \varphi_i \cdot \det D(\alpha_i^{-1} \circ \beta) \right)}_{>0} dx^1 \wedge \dots \wedge dx^d$$

We then let $p \in M$ with β a patch around p . Without loss of generality assume β is positive, then $\beta^* \omega = \underbrace{u}_{>0} dx^1 \wedge \dots \wedge dx^d \quad \square$

Proposition 6.5.5. *Suppose $\omega_1, \omega_2 \in \Omega^d(M)$ induce the same orientation. Then $\exists f \in C^\infty(M)$ such that $f(p) > 0 \forall p \in M$ such that $\omega_1 = f \cdot \omega_2$*

Proof: f is uniquely defined by $\omega_1 = f \cdot \omega_2$ since ω_2 is nowhere vanishing. Let α be a positive chart with respect to the induced orientation, then:

$$\begin{aligned} \alpha^* \omega_1 &= \alpha^* f \cdot \alpha^* \omega_2 \\ \iff u_1 dx^1 \wedge \dots \wedge dx^d &= \alpha^* f \cdot u_2 dx^1 \wedge \dots \wedge dx^d \\ \implies \underbrace{u_1}_{>0} &= \alpha^* f \cdot \underbrace{u_2}_{>0} \implies f > 0 \end{aligned}$$

Theorem 6.5.6. *There is a one-to-one correspondence between volume forms (up to multiplication by some positive function) and orientations on M*

Proof: Combination of the previous 3 theorems.

Example 6.5.7.

$$M = S^1, \quad \omega = x \cdot dy - y \cdot dx$$

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

$$T_{(x,y)} S^1 = \ker Df(p) = \{(w, u) \in \mathbb{R}^2 \mid 2xu + 2yv = 0\} \text{ i.e } (-y, x) \in T_{(x,y)} S^1.$$

$$\omega((-y, x)) = x^2 + y^2 = 1$$

Thus ω is a nowhere vanishing, thus a volume form.

Proposition 6.5.8. *Let ω_M, ω_N be volume forms inducing orientations on M and N respectively. Then f is orientation preserving if and only if $f^* \omega_N = u \cdot \omega_M$ for some $u > 0$.*

Corollary 6.5.9. *Let $f : S^n \rightarrow S^n$ with $f(x) = -x$ where S^n is the unit n -sphere. f is orientation preserving if and only if n is odd.*

Proof: f extends to $F : D^{n+1} \rightarrow D^{n+1}$, $F(x) = -x$ where D^{n+1} is the unit $(n+1)$ ball. Let α be the standard orientation on D^{n+1} . Then $\{F \circ \alpha\}$ has the sign of $\text{sign}(\det DF) \cdot (\text{sign}(\{\alpha\}))$. $\text{sign}(\det DF) = (-1)^{n+1}$, thus, the restricted orientation differ by $(-1)^{n+1}$.

Chapter 7

Integrating Forms

7.1 Definition of Integration of Forms

Recall that if $\varphi : \mathcal{U}_1 \rightarrow \mathcal{U}_2$ is a diffeomorphism and orientation preserving, then:

$$\int_{\mathcal{U}_1} \varphi^* \omega = \int_{\mathcal{U}_2} \omega, \forall \omega \in \Omega^d(\mathcal{U}_2)$$

Definition 7.1.1 (Integral of a form).

Let M be an orientated manifold with boundary.

$$\int_M \omega \text{ is defined as follows:}$$

Given positive charts $\{\alpha_i, \mathcal{U}_i\}$, we choose a partition of unity $\{\varphi_i\}$ subordinate to $\alpha_i(\mathcal{U}_i)$, then:

$$\int_M \omega = \sum_i \int_{\mathcal{U}_i} \alpha_i^*(\varphi_i \cdot \omega)$$

Remark 7.1.2.

We restrict ourselves to the case $\omega \in \Omega_c^d(M)$ (the differential forms with compact support). Note if M is compact, then $\Omega^d(M) = \Omega_c^d(M)$

Proposition 7.1.3. *This is well-defined, i.e it is independent of our choice of α_i, φ_i*

Proof: Choose $\{\alpha_i\}$ to be the collection of all positive charts. It then suffices to prove our

choice of partition of unity does not matter. Let $\{\varphi_i\}, \{\psi_j\}$ be two partitions of unity:

$$\begin{aligned}
\sum_i \int_{U_i} \alpha_i^*(\varphi_i \omega) &= \sum_i \int \alpha_i^* \left(\sum_j \varphi_i \psi_j \omega \right) \\
&= \sum_{i,j} \int_{U_i} \underbrace{\alpha_i^*}_{(\alpha_j^{-1} \alpha_i)^* \alpha_j^*} (\varphi_i \psi_j \omega) \\
&\quad (\alpha_j^{-1} \alpha_i) \text{ is a diffeomorphism} \\
&= \sum_{i,j} \int_{\alpha_i^{-1}(\alpha_i(U_i) \cap \alpha_j(U_j))} (\alpha_j^{-1} \alpha_i)^* \alpha_j^* (\varphi_i \psi_j \omega) \\
&= \sum_{i,j} \int_{\alpha_i^{-1}(\alpha_i(U_i) \cap \alpha_j(U_j))} \alpha_j^* (\varphi_i \psi_j \omega) = \dots = \sum_j \int \alpha_j^* (\psi_j \omega)
\end{aligned}$$

Example 7.1.4.

$M = S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ with positive charts $\alpha_1(0, 2\pi) \rightarrow S^1 \setminus \{(0, 1)\}, t \mapsto (\cos(t), \sin(t))$ and $\alpha_2(-\pi, \pi) \rightarrow S^1 \setminus \{(-1, 0)\}, t \mapsto (\cos(t), \sin(t))$. We claim that α_i are positive patches. Take $\omega = f(x, y)dx + g(x, y)dy$. Let φ_1, φ_2 be a partition of unity subordinate to $\text{Im}(\alpha_i)$

$$\begin{aligned}
\int_{S^1} \omega &= \int_0^{2\pi} \alpha_1^*(\varphi_1 \omega) + \int_0^{2\pi} \alpha_2^*(\varphi_2 \omega) \\
\alpha_1^*(\varphi_1 \omega) &= \varphi_1(\cos(t), \sin(t)) (f(\cos(t), \sin(t))d(\cos(t)) + g(\cos(t), \sin(t))d(\sin(t))) \\
&= \int_0^\pi \alpha_1^*(\varphi_1 \omega) + \int_\pi^{2\pi} \alpha_1^*(\varphi_1 \omega) + \int_0^\pi \alpha_2^*(\varphi_2 \omega) + \int_\pi^{2\pi} \alpha_2^*(\varphi_2 \omega)
\end{aligned}$$

But $\alpha_1|_{(0,\pi)} = \alpha_2|_{(0,\pi)}$ and $\alpha_1|_{(\pi,2\pi)} = \alpha_2|_{(\pi,2\pi)} \circ t$ such that $t(x) = x + 2\pi$

$$\begin{aligned}
&\int_0^\pi \alpha_1^*((\varphi_1 + \varphi_2)\omega) + \int_\pi^{2\pi} \alpha_1^*(\varphi_1 + \varphi_2)\omega = \int_0^{2\pi} \alpha_1^* \omega \\
&\int_0^{2\pi} f(\cos(t), \sin(t))d(\cos(t)) + g(\cos(t), \sin(t))d(\sin(t))
\end{aligned}$$

Theorem 7.1.5. Let M be a manifold that is diffeomorphic to S^1 and $\alpha: (0, 2\pi) \rightarrow M$ is a positive chart such that $\alpha(0, 2\pi) = M \setminus \{p\}$ for some $p \in M$, then

$$\int_0^{2\pi} \alpha^* \omega = \int_M \omega$$

Theorem 7.1.6. The integral satisfies:

1. *Linearity:* $\int_M \alpha \cdot \omega + \beta \cdot \eta = \alpha \cdot \int_M \omega + \beta \cdot \int_M \eta$
2. If $-M$ denotes M with opposite orientation, then $\int_M \Omega = -\int_{-M} \omega$

Theorem 7.1.7. *Let $F : M \rightarrow N$ be orientation-preserving diffeomorphism. Then $\int_M F^* \omega = \int_N \omega$ for $\Omega_c^d(M)$*

Proof: Let $\{\varphi_i\}$ be a partition of unity subordinate to positive charts $\{\alpha_i\}$ covering M . Then $\{(F^{-1})^* \varphi_i\}, \{F \circ \alpha_i\}$ is the same data on N .

$$\begin{aligned} \int_N \omega &= \sum_i \int_{U_i} (F \circ \alpha_i)^* ((F^{-1})^* \varphi_i \cdot \omega) \\ &= \sum_i \int_{U_i} \alpha_i^* (\varphi_i \cdot F^* \omega) \\ &= \int_M F^* \omega \end{aligned}$$

7.2 Stoke's Theorem

"When you are face to face with a difficulty, you are up against a discovery."

-William Thompson (Lord Kelvin)

Before we look at "Stoke's" Theorem, a short historical aside. This theorem was not in fact discovered by George Stokes but William Thompson. It attained the name "Stoke's Theorem" because of his habit of putting it on examinations at Cambridge before it was widely known. Hopefully your exam is less diabolical.

Lemma 7.2.1. *Let $b : \mathbb{R}^{d-1} \hookrightarrow \mathbb{H}^d$ be the inclusion of the boundary. For any $\eta \in \Omega_c^d(\mathbb{H}^d)$, we have:*

$$\int_{\mathbb{H}^d} d\eta = (-1)^d \int_{\mathbb{R}^{d-1}} b^* \eta$$

Proof: η is of the form: $\eta = \sum_i f_i dx^{I_i}$. Both sides are linear in η , we can thus assume $\eta = f \cdot dx^{I_i}$ where I_i is the ascending index not including the i th term.

$$d\eta = df \wedge dx^{I_i} = \frac{\partial f}{\partial x^i} (-1)^{i-1} dx^1 \wedge \dots \wedge dx^d$$

Since f has compact support, we can assume that $\text{supp}(f) \subset [-L, L]^{d-1} \times [0, L]$ and $f(x^1, \dots, x^d) = 0$ whenever $x^i = \pm L$ for $i \neq d$ or $x^d = L$.

$$\int_{\mathbb{H}^d} d\eta = \int_{[-L, L]^{d-1} \times [0, L]} \frac{\partial f}{\partial x^i} (-1)^{i-1} dx^1 \wedge \dots \wedge dx^d$$

We consider the 2 cases for the value of this integral:

$$(i \neq d) \int_{[-L,L]^{d-2} \times [0,L]} (-1)^{i-1} \underbrace{\left(\int_{[-L,L]} \frac{\partial f}{\partial x^i} dx^i \right)}_{f(x^1, \dots, L, \dots, x^d) - f(x^1, \dots, -L, \dots, x^d)} dx^{I_i} = 0$$

$$(i = d) \int_{[-L,L]^{d-1}} (-1)^{i-1} \underbrace{\left(\int_{[0,L]} \frac{\partial f}{\partial x^d} dx^d \right)}_{f(x^1, \dots, x^{d-1}, L) - f(x^1, \dots, x^{d-1}, 0)} dx^{I_d} = \int_{[-L,L]^{d-1}} (-1)^d f(x^1, \dots, x^{d-1}, 0) dx^{I_d}$$

On the other side we have:

$$b^* \eta = \begin{cases} i \neq d & 0 \\ i = d & b^* f \cdot dx^{I_d} \end{cases}, \quad \int_{\mathbb{R}^{d-1}} b^* \eta = \begin{cases} i \neq d & 0 \\ i = d & \int_{[-L,L]^{d-1}} (-1)^d f(x^1, \dots, x^{d-1}, 0) dx^{I_d} \end{cases}$$

Remark 7.2.2.

Let $b : \partial M \hookrightarrow M$ be the inclusion of the boundary. Suppose M is orientated. For $\omega \in \Omega^{d-1}(M)$ we write:

$$\int_{\partial M} \omega = \int_{\partial M} b^* \omega$$

This integral is computed in the induced orientation.

Corollary 7.2.3.

$$\int_{\mathbb{R}^d} d\eta = 0, \quad \eta \in \Omega_c^{d-1}(\mathbb{R})$$

Theorem 7.2.4 (Stoke's Theorem). *Let M be an orientated manifold with boundary. Then:*

$$\int_M d\omega = \int_{\partial M} \omega, \quad \omega \in \Omega_c^{d-1}(M)$$

Remark 7.2.5.

- If M is compact, then $\Omega^*(M) = \Omega_c^*(M)$
- If M has no boundary, then $\int d\omega = 0$

Proof: Choose a partition of unity subordinate to a covering of M by positive charts.

$$\omega = \sum_i \varphi_i \omega$$

Since $\text{supp}(\omega)$ is compact, only finitely many $\varphi_i \omega$ are non-zero. $\varphi_i \omega$ has its support contained in a single positive patch. By linearity, we can assume ω has this property. Let $\alpha : U \rightarrow V$ be that chart. $U \subset \mathbb{H}^d$

$$\int_M d\omega = \int_{\mathbb{H}^d} \alpha^* d\omega = \int_{\mathbb{H}^d} d\alpha^* \omega = (-1)^d \int_{\mathbb{R}^{d-1}} \underbrace{b_{\mathbb{H}^d}^* \alpha^*}_{=\alpha \circ b_{\mathbb{H}^d}} \omega$$

But $\alpha \circ \mathbf{b}_{\mathbb{H}^d} : \mathbf{b}_{\mathbb{H}^d}^{-1}(\mathcal{U}) \rightarrow V$ is the restricted chart on ∂M and $\mathbf{b}^* \omega$ has support contained in the image. More precisely, $\exists$ a chart $\bar{\alpha} : \mathbf{b}_{\mathbb{H}^d}^{-1}(\mathcal{U}) \rightarrow \mathbb{R}^{d-1}$ such that $\mathbf{b}_M \circ \bar{\alpha} = \alpha \circ \mathbf{b}_{\mathbb{H}^d}$. Let ∂M^{res} be ∂M with the restricted orientation such that $\alpha \circ \mathbf{b}$ is a positive chart.

$$\begin{aligned} \int_{\partial M} \mathbf{b}_M^* \omega &= (-1)^d \int_{\partial M^{\text{res}}} \mathbf{b}_M^* \omega = (-1)^d \int_{\mathbb{R}^{d-1}} \underbrace{\bar{\alpha}^* \mathbf{b}_M^* \omega}_{(\mathbf{b}_M \circ \bar{\alpha})^* \omega} \\ &= (-1)^d \int_{\mathbb{R}^{d-1}} (\alpha \circ \mathbf{b}_{\mathbb{H}^d})^* \omega = (-1)^d \int_{\mathbb{R}^{d-1}} \mathbf{b}_{\mathbb{H}^d}^* \alpha^* \omega \quad \square \end{aligned}$$

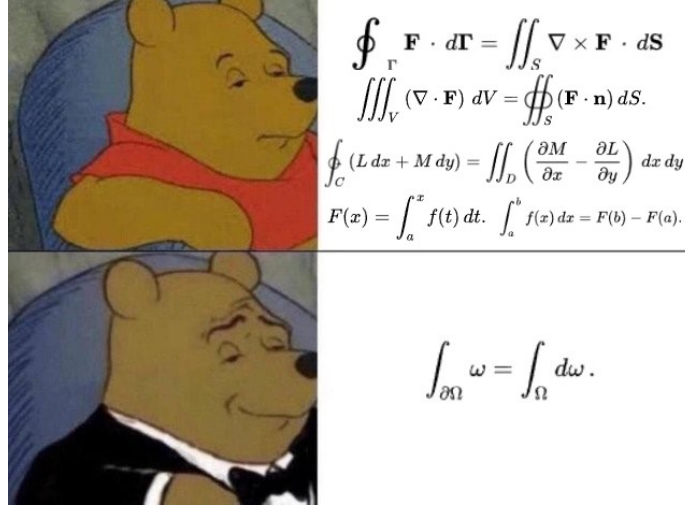


Figure 7.1: Stoke's theorem generalises many of our favourite results

Remark 7.2.6.

If M is a 1-manifold with boundary, an orientation on ∂M is $\sigma : \partial M \rightarrow \{\pm 1\}$. We define it as follows:

If $\exists$ a positive patch $\alpha : [0, \varepsilon) \rightarrow M$ then $\sigma(\alpha(0)) = -1$. If $\alpha : (-\varepsilon, 0] \rightarrow M$ then $\sigma(\alpha(0)) = 1$.

If we define:

$$\int_{\partial M} f = \sum_{p \in \partial M} \sigma(p) \cdot f(p)$$

then Stoke's theorem holds (with essentially the same proof). For $M = [0, 1]$, it says:

$$\int_0^1 \frac{\partial f}{\partial x} \cdot dx = \int_M df = \int_{\partial M} f = f(1) - f(0)$$

Example 7.2.7.

$f(x, y) = x^2 + y^2 - 1$, $M = \{(x, y) \in \mathbb{R}^2 \mid f(x, y) \leq 0\}$, $\partial M = S^1$.

$$\int_M dx \wedge dy = \int_M 1 \cdot dx \cdot dy = \pi$$

$$\int_M dx \wedge dy = \int_M d(x \cdot dy) = \int_{\partial M} x \cdot dy = \int_0^{2\pi} \cos^2(t) \cdot dt = \pi$$